

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
MTD selection (%)						
U2 - shared alpha	63.9	15.1	7.9	3.5	0.7	8.9
U2 - dose-specific alpha	63	15.7	8	3.4	0.9	9
U3 - shared alpha	63.1	16.1	7.3	4	0.6	8.9
U3 - dose-specific alpha	63.9	15.1	8.4	3	0.8	8.8
OBD selection (%)						
U2 - shared alpha	15.9	8	1.6	0.5	0.2	73.8
U2 - dose-specific alpha	16.1	8	2.1	0.5	0.3	73
U3 - shared alpha	17.3	7.7	1.3	0.3	0.2	73.2
U3 - dose-specific alpha	18.5	7.2	1.1	0.2	0.2	72.8
Mean administrations						
U2 - shared alpha	11.1	3.6	2.1	0.9	0.3	-
U2 - dose-specific alpha	11.1	3.5	2.1	0.9	0.4	-
U3 - shared alpha	11.1	3.6	2.2	0.9	0.3	-
U3 - dose-specific alpha	11.1	3.6	2.2	0.9	0.3	-
Mean IPDE administrations						
U2 - shared alpha	3.2	1.8	0.7	0.4	0.1	-
U2 - dose-specific alpha	3.2	1.8	0.7	0.4	0.2	-
U3 - shared alpha	3.2	1.8	0.7	0.4	0.1	-
U3 - dose-specific alpha	3.2	1.7	0.8	0.4	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						642.5
U2 - dose-specific alpha						643.7
U3 - shared alpha						645
U3 - dose-specific alpha						644.9

Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
MTD selection (%)						
U2 - shared alpha	22.4	16.7	29.2	20.8	10.4	0.5
U2 - dose-specific alpha	22.3	17.4	29.2	20.5	10.1	0.5
U3 - shared alpha	22.4	16.5	29.5	20.4	10.7	0.5
U3 - dose-specific alpha	22.2	17.1	28.6	21.2	10.4	0.5
OBD selection (%)						
U2 - shared alpha	31.7	15	33.2	11.2	3.7	5.2
U2 - dose-specific alpha	31.4	15.4	32	12.4	3.7	5.1
U3 - shared alpha	34.3	16.3	31.2	10.2	3	5
U3 - dose-specific alpha	34.4	16.8	29.8	11.5	2.5	5
Mean administrations						
U2 - shared alpha	10.8	5.1	7.2	3.8	2.2	-
U2 - dose-specific alpha	10.9	5.2	7.2	3.8	2.1	-
U3 - shared alpha	10.8	5.1	7.1	3.9	2.1	-
U3 - dose-specific alpha	10.8	5.1	7.2	3.9	2.2	-
Mean IPDE administrations						
U2 - shared alpha	3.1	2.4	1.9	1	0.5	-
U2 - dose-specific alpha	3.1	2.5	1.9	1	0.5	-
U3 - shared alpha	3.1	2.4	1.9	1.1	0.5	-
U3 - dose-specific alpha	3.1	2.4	1.9	1.1	0.6	-
Mean trial duration (days)						Days
U2 - shared alpha						1107.9
U2 - dose-specific alpha						1107
U3 - shared alpha						1107.9
U3 - dose-specific alpha						1107.5

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
MTD selection (%)						
U2 - shared alpha	22.4	16.9	29.9	19.4	10.9	0.5
U2 - dose-specific alpha	22.7	17.1	28.6	20.3	10.8	0.5
U3 - shared alpha	22.6	16.5	29.8	20	10.6	0.5
U3 - dose-specific alpha	22.7	16.8	29.2	19.9	10.9	0.5
OBD selection (%)						
U2 - shared alpha	10.5	39.5	21	8.2	2.7	18.1
U2 - dose-specific alpha	10.2	39.2	22.2	7.9	2.2	18.3
U3 - shared alpha	12.6	42.7	18.9	5.7	1.9	18.2
U3 - dose-specific alpha	12	41.6	19.1	7.1	2	18.2
Mean administrations						
U2 - shared alpha	7.8	7.5	6.1	3.6	2	-
U2 - dose-specific alpha	7.8	7.5	6.1	3.6	2	-
U3 - shared alpha	7.8	7.6	6	3.6	2	-
U3 - dose-specific alpha	7.8	7.5	6.1	3.6	2	-
Mean IPDE administrations						
U2 - shared alpha	2	3.3	1.7	1.1	0.6	-
U2 - dose-specific alpha	2	3.3	1.7	1.1	0.6	-
U3 - shared alpha	2	3.3	1.7	1.1	0.6	-
U3 - dose-specific alpha	2	3.3	1.7	1.1	0.6	-
Mean trial duration (days)						Days
U2 - shared alpha						1011.7
U2 - dose-specific alpha						1011.3
U3 - shared alpha						1011.7
U3 - dose-specific alpha						1011.1

Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
MTD selection (%)						
U2 - shared alpha	2.1	2.6	20	38.4	36.9	0
U2 - dose-specific alpha	2.1	2.7	20.1	37.8	37.3	0
U3 - shared alpha	2.1	2.9	19.7	38.4	36.9	0
U3 - dose-specific alpha	2.1	2.6	20.3	38	37	0
OBD selection (%)						
U2 - shared alpha	6.4	7.2	21.6	42.5	19.4	2.9
U2 - dose-specific alpha	6.1	6.5	21.6	42.4	20.6	2.8
U3 - shared alpha	10.6	8.8	18.7	41.8	17.2	2.9
U3 - dose-specific alpha	10.7	8.1	18.7	41.5	18.1	2.9
Mean administrations						
U2 - shared alpha	4.3	4.2	6.5	8	6.5	-
U2 - dose-specific alpha	4.3	4.1	6.5	8	6.5	-
U3 - shared alpha	4.3	4.1	6.4	8	6.5	-
U3 - dose-specific alpha	4.3	4.2	6.5	8	6.5	-
Mean IPDE administrations						
U2 - shared alpha	0.6	2.7	2	2.6	1.5	-
U2 - dose-specific alpha	0.6	2.7	2	2.6	1.5	-
U3 - shared alpha	0.6	2.7	2	2.6	1.5	-
U3 - dose-specific alpha	0.6	2.7	2	2.6	1.5	-
Mean trial duration (days)						Days
U2 - shared alpha						1097.8
U2 - dose-specific alpha						1097
U3 - shared alpha						1097.3
U3 - dose-specific alpha						1098.4

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
MTD selection (%)						
U2 - shared alpha	5.6	3.1	12	22.1	57.1	0.1
U2 - dose-specific alpha	5.6	3.2	11.8	21.3	58	0.1
U3 - shared alpha	5.6	3.1	11.6	22.2	57.4	0.1
U3 - dose-specific alpha	5.6	3	12	22.2	57.1	0.1
OBD selection (%)						
U2 - shared alpha	3.7	36.3	24.8	18.5	11.2	5.5
U2 - dose-specific alpha	3.9	34.8	25.9	19.3	10.6	5.5
U3 - shared alpha	6.9	42.5	22.5	16.3	6.3	5.5
U3 - dose-specific alpha	7.6	41.7	23.8	15	6.4	5.5
Mean administrations						
U2 - shared alpha	4.5	6.5	6.5	5.8	5.6	-
U2 - dose-specific alpha	4.5	6.4	6.6	5.8	5.7	-
U3 - shared alpha	4.5	6.5	6.6	5.7	5.6	-
U3 - dose-specific alpha	4.5	6.5	6.6	5.7	5.7	-
Mean IPDE administrations						
U2 - shared alpha	0.7	3.5	2.1	2.2	1.9	-
U2 - dose-specific alpha	0.7	3.4	2.1	2.2	2	-
U3 - shared alpha	0.7	3.4	2.1	2.2	1.9	-
U3 - dose-specific alpha	0.7	3.5	2.1	2.2	2	-
Mean trial duration (days)						Days
U2 - shared alpha						1021.1
U2 - dose-specific alpha						1021.1
U3 - shared alpha						1020.3
U3 - dose-specific alpha						1020.6

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
MTD selection (%)						
U2 - shared alpha	56.6	18.1	8.9	4.3	1	11.1
U2 - dose-specific alpha	56.3	17.6	9.2	4.8	0.9	11.2
U3 - shared alpha	56.1	17.7	9.5	4.6	0.8	11.3
U3 - dose-specific alpha	56.6	18	9	4.6	1	10.8
OBD selection (%)						
U2 - shared alpha	66.7	13	3.6	1.3	0.4	15
U2 - dose-specific alpha	65.7	13.6	3.9	1.4	0.4	15
U3 - shared alpha	67.3	13.2	3.3	1	0.4	14.8
U3 - dose-specific alpha	67.7	12.8	3.1	1	0.4	15
Mean administrations						
U2 - shared alpha	19.2	4	2.5	1.1	0.4	-
U2 - dose-specific alpha	19.2	4	2.5	1.1	0.4	-
U3 - shared alpha	19.1	4	2.6	1.1	0.4	-
U3 - dose-specific alpha	19.2	3.9	2.5	1.1	0.4	-
Mean IPDE administrations						
U2 - shared alpha	5.2	1.5	0.7	0.4	0.1	-
U2 - dose-specific alpha	5.3	1.6	0.7	0.4	0.1	-
U3 - shared alpha	5.2	1.6	0.8	0.4	0.1	-
U3 - dose-specific alpha	5.3	1.5	0.7	0.4	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1056.7
U2 - dose-specific alpha						1056.6
U3 - shared alpha						1057.1
U3 - dose-specific alpha						1056.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

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Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
MTD selection (%)						
U2 - shared alpha	22.8	14	21	23	18.5	0.7
U2 - dose-specific alpha	22.9	14.3	21.2	22.9	18	0.7
U3 - shared alpha	22.8	14.1	21.6	22.9	17.9	0.7
U3 - dose-specific alpha	22.6	14.6	21.1	22.7	18.3	0.7
OBD selection (%)						
U2 - shared alpha	22.1	37.1	20.3	11	3.6	5.9
U2 - dose-specific alpha	23.4	36.4	20	10.4	3.9	5.9
U3 - shared alpha	25.1	39.8	18.1	8.5	2.7	5.8
U3 - dose-specific alpha	25.6	39.9	17.2	8.8	2.6	5.9
Mean administrations						
U2 - shared alpha	10.1	7	5.5	3.8	2.5	-
U2 - dose-specific alpha	10.1	7	5.6	3.8	2.5	-
U3 - shared alpha	10.1	7	5.6	3.8	2.5	-
U3 - dose-specific alpha	10.2	7.1	5.5	3.7	2.5	-
Mean IPDE administrations						
U2 - shared alpha	2.8	2.8	1.4	1.1	0.7	-
U2 - dose-specific alpha	2.8	2.8	1.4	1.1	0.7	-
U3 - shared alpha	2.8	2.8	1.4	1.1	0.7	-
U3 - dose-specific alpha	2.8	2.8	1.3	1.1	0.7	-
Mean trial duration (days)						Days
U2 - shared alpha						1118.8
U2 - dose-specific alpha						1119.2
U3 - shared alpha						1119.8
U3 - dose-specific alpha						1118.7

Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
MTD selection (%)						
U2 - shared alpha	65.8	11.8	6.8	2	0.6	13
U2 - dose-specific alpha	65.5	12.7	5.9	2	0.6	13.3
U3 - shared alpha	65.6	12.6	6	2.3	0.5	13
U3 - dose-specific alpha	65.7	12.4	6.4	2.1	0.7	12.7
OBD selection (%)						
U2 - shared alpha	26.2	8.5	3.2	1	0.4	60.7
U2 - dose-specific alpha	26.2	9.1	3	1	0.3	60.4
U3 - shared alpha	27.4	8.8	2.3	0.9	0.3	60.3
U3 - dose-specific alpha	27.4	8.4	2.5	1.1	0.3	60.3
Mean administrations						
U2 - shared alpha	13.4	3.6	1.9	0.8	0.3	-
U2 - dose-specific alpha	13.4	3.6	1.9	0.8	0.3	-
U3 - shared alpha	13.4	3.6	1.9	0.8	0.3	-
U3 - dose-specific alpha	13.4	3.6	1.9	0.8	0.3	-
Mean IPDE administrations						
U2 - shared alpha	3.9	1.6	0.6	0.3	0.1	-
U2 - dose-specific alpha	3.9	1.6	0.6	0.3	0.1	-
U3 - shared alpha	3.9	1.7	0.6	0.3	0.1	-
U3 - dose-specific alpha	3.9	1.7	0.6	0.3	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						733.4
U2 - dose-specific alpha						733.6
U3 - shared alpha						732.9
U3 - dose-specific alpha						734.8

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
MTD selection (%)						
U2 - shared alpha	22.8	8.1	15.7	17.7	35.2	0.5
U2 - dose-specific alpha	22.9	8.1	15.5	18.5	34.5	0.5
U3 - shared alpha	22.8	8	15.5	18.7	34.5	0.5
U3 - dose-specific alpha	22.9	8	15.4	18.3	34.9	0.5
OBD selection (%)						
U2 - shared alpha	7.7	8.6	17.8	13.6	26.4	25.9
U2 - dose-specific alpha	8.7	8	17.4	14.6	25.1	26.2
U3 - shared alpha	13.9	9.8	15.1	12.7	22.5	26
U3 - dose-specific alpha	13.7	10.1	15.3	11.8	22.7	26.4
Mean administrations						
U2 - shared alpha	6.9	4.1	5.3	4.1	5.2	-
U2 - dose-specific alpha	6.9	4.1	5.3	4	5.3	-
U3 - shared alpha	6.8	4.1	5.3	4.1	5.3	-
U3 - dose-specific alpha	6.9	4.1	5.3	4	5.3	-
Mean IPDE administrations						
U2 - shared alpha	1.7	2.4	1.8	1.6	1.5	-
U2 - dose-specific alpha	1.7	2.4	1.8	1.6	1.5	-
U3 - shared alpha	1.7	2.4	1.8	1.6	1.5	-
U3 - dose-specific alpha	1.7	2.4	1.8	1.6	1.5	-
Mean trial duration (days)						Days
U2 - shared alpha						908.6
U2 - dose-specific alpha						907.7
U3 - shared alpha						908.7
U3 - dose-specific alpha						907

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
MTD selection (%)						
U2 - shared alpha	47	18.5	15.9	10.6	4.9	3.1
U2 - dose-specific alpha	46.8	17.8	16.7	10.3	5.3	3.1
U3 - shared alpha	46.9	18	16.1	10.6	5.2	3.2
U3 - dose-specific alpha	47.2	17.6	16.4	9.7	5.9	3.2
OBD selection (%)						
U2 - shared alpha	6.4	29.1	12	4.3	1.9	46.3
U2 - dose-specific alpha	6.4	27.6	13	4.4	2.3	46.3
U3 - shared alpha	6.9	30.5	10.7	4	1.6	46.3
U3 - dose-specific alpha	7.6	29.8	10.6	3.9	1.8	46.3
Mean administrations						
U2 - shared alpha	8.4	5.9	4.1	2.2	1.1	-
U2 - dose-specific alpha	8.4	5.8	4.1	2.2	1.2	-
U3 - shared alpha	8.4	6	4.1	2.2	1.2	-
U3 - dose-specific alpha	8.4	5.8	4.1	2.1	1.2	-
Mean IPDE administrations						
U2 - shared alpha	2.2	2.5	1.1	0.7	0.3	-
U2 - dose-specific alpha	2.2	2.5	1.1	0.7	0.3	-
U3 - shared alpha	2.2	2.5	1.1	0.6	0.3	-
U3 - dose-specific alpha	2.2	2.5	1.1	0.6	0.3	-
Mean trial duration (days)						Days
U2 - shared alpha						817.3
U2 - dose-specific alpha						815.1
U3 - shared alpha						817.3
U3 - dose-specific alpha						816

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

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Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
MTD selection (%)						
U2 - shared alpha	47.7	30.8	15.4	2.7	0.2	3.2
U2 - dose-specific alpha	47.7	31	15.1	2.8	0	3.4
U3 - shared alpha	47.6	30.9	15.4	2.9	0	3.2
U3 - dose-specific alpha	47.3	31.6	15.3	2.6	0	3.2
OBD selection (%)						
U2 - shared alpha	58.1	20.8	4.6	1	0.1	15.4
U2 - dose-specific alpha	57.9	20.6	5.5	0.8	0	15.2
U3 - shared alpha	60.4	19.2	4.4	0.6	0	15.4
U3 - dose-specific alpha	62.1	18.5	3.3	0.7	0	15.4
Mean administrations						
U2 - shared alpha	16.8	5.9	3.5	1.1	0.2	-
U2 - dose-specific alpha	17	5.7	3.5	1.1	0.2	-
U3 - shared alpha	16.9	5.7	3.6	1	0.2	-
U3 - dose-specific alpha	16.9	5.8	3.6	1	0.2	-
Mean IPDE administrations						
U2 - shared alpha	5.1	2.5	1.1	0.4	0.1	-
U2 - dose-specific alpha	5.2	2.5	1.1	0.4	0.1	-
U3 - shared alpha	5.2	2.5	1.1	0.4	0.1	-
U3 - dose-specific alpha	5.1	2.5	1.1	0.4	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1003.3
U2 - dose-specific alpha						1001.8
U3 - shared alpha						1002.3
U3 - dose-specific alpha						1002.4

Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
MTD selection (%)						
U2 - shared alpha	3.9	2.5	18.8	43.2	31.5	0.1
U2 - dose-specific alpha	3.9	2.1	19.1	41.9	32.9	0.1
U3 - shared alpha	3.9	2.4	19.3	42.2	32.1	0.1
U3 - dose-specific alpha	3.9	2.4	19	42.5	32.1	0.1
OBD selection (%)						
U2 - shared alpha	12.2	11.4	9.6	37.6	23.8	5.4
U2 - dose-specific alpha	12	11.6	10.1	36.7	24.3	5.3
U3 - shared alpha	18.8	11.5	7.7	34.7	21.8	5.5
U3 - dose-specific alpha	19.3	11.6	6.8	34.5	22.5	5.3
Mean administrations						
U2 - shared alpha	4.7	4.5	5.2	7.5	7.3	-
U2 - dose-specific alpha	4.7	4.5	5.2	7.3	7.4	-
U3 - shared alpha	4.7	4.5	5.3	7.4	7.2	-
U3 - dose-specific alpha	4.7	4.5	5.2	7.4	7.3	-
Mean IPDE administrations						
U2 - shared alpha	0.8	2.8	1.6	2.8	1.6	-
U2 - dose-specific alpha	0.8	2.9	1.6	2.7	1.6	-
U3 - shared alpha	0.8	2.9	1.7	2.7	1.6	-
U3 - dose-specific alpha	0.8	2.8	1.7	2.7	1.6	-
Mean trial duration (days)						Days
U2 - shared alpha						1074.5
U2 - dose-specific alpha						1076.2
U3 - shared alpha						1074
U3 - dose-specific alpha						1075.7

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
MTD selection (%)						
U2 - shared alpha	42.6	18.8	19.9	11.2	5.3	2.2
U2 - dose-specific alpha	42.9	18.6	19.7	11	5.6	2.2
U3 - shared alpha	42.7	18.7	19.7	12	4.8	2.1
U3 - dose-specific alpha	43	18.2	19.8	11.5	5.3	2.2
OBD selection (%)						
U2 - shared alpha	8.3	29.4	13.8	5.1	2	41.4
U2 - dose-specific alpha	8.3	29.3	14.3	4.7	2	41.4
U3 - shared alpha	9.5	30.2	12.9	4.6	1.3	41.5
U3 - dose-specific alpha	9.8	30.7	12.6	3.8	1.5	41.6
Mean administrations						
U2 - shared alpha	8.2	6.2	4.4	2.4	1.2	-
U2 - dose-specific alpha	8.2	6.2	4.4	2.4	1.3	-
U3 - shared alpha	8.2	6.2	4.5	2.4	1.2	-
U3 - dose-specific alpha	8.2	6.1	4.5	2.4	1.2	-
Mean IPDE administrations						
U2 - shared alpha	2.2	2.8	1.3	0.8	0.4	-
U2 - dose-specific alpha	2.1	2.8	1.3	0.8	0.4	-
U3 - shared alpha	2.1	2.8	1.4	0.8	0.4	-
U3 - dose-specific alpha	2.1	2.8	1.4	0.8	0.4	-
Mean trial duration (days)						Days
U2 - shared alpha						822
U2 - dose-specific alpha						824.2
U3 - shared alpha						822.3
U3 - dose-specific alpha						822.3

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
MTD selection (%)						
U2 - shared alpha	9.3	6.1	22.7	30.2	31.5	0.2
U2 - dose-specific alpha	9.4	5.7	22.9	30.8	31	0.2
U3 - shared alpha	9.3	6.6	22.3	30.2	31.4	0.2
U3 - dose-specific alpha	9.4	6.1	22.2	31.1	31	0.2
OBD selection (%)						
U2 - shared alpha	16.2	11.7	13.7	29	16.6	12.8
U2 - dose-specific alpha	15.7	12.6	13.3	28.8	17.3	12.3
U3 - shared alpha	24.8	14.3	9.8	26.3	11.9	12.9
U3 - dose-specific alpha	24.3	15.2	9.5	25.4	13.1	12.5
Mean administrations						
U2 - shared alpha	5.8	5.1	5.7	6.2	5.2	-
U2 - dose-specific alpha	5.9	5	5.8	6.2	5.2	-
U3 - shared alpha	5.9	5.1	5.7	6.2	5.2	-
U3 - dose-specific alpha	5.9	5.1	5.7	6.2	5.2	-
Mean IPDE administrations						
U2 - shared alpha	1.3	3	2	2.4	1.5	-
U2 - dose-specific alpha	1.3	2.9	2	2.4	1.5	-
U3 - shared alpha	1.3	3	1.9	2.4	1.5	-
U3 - dose-specific alpha	1.3	3	2	2.4	1.5	-
Mean trial duration (days)						Days
U2 - shared alpha						984.3
U2 - dose-specific alpha						985.5
U3 - shared alpha						984.3
U3 - dose-specific alpha						984.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
MTD selection (%)						
U2 - shared alpha	17.4	19	33	19.9	10.3	0.4
U2 - dose-specific alpha	17.3	18	33.2	20.8	10.3	0.4
U3 - shared alpha	17.5	18.4	32.9	20.8	10	0.4
U3 - dose-specific alpha	17.4	19	32.9	20.1	10.2	0.4
OBD selection (%)						
U2 - shared alpha	43.2	23.6	11.2	3.2	0.9	17.9
U2 - dose-specific alpha	44.1	22.9	10.4	3.7	0.7	18.2
U3 - shared alpha	52.3	20.1	6.9	2.3	0.4	18
U3 - dose-specific alpha	53.4	20.1	6.7	1.3	0.5	18
Mean administrations						
U2 - shared alpha	10	6.5	5.8	3.3	1.7	-
U2 - dose-specific alpha	10	6.6	5.8	3.3	1.7	-
U3 - shared alpha	10	6.6	5.8	3.3	1.7	-
U3 - dose-specific alpha	10.1	6.6	5.8	3.3	1.7	-
Mean IPDE administrations						
U2 - shared alpha	3	3.4	2	1.4	0.7	-
U2 - dose-specific alpha	3	3.4	2	1.4	0.7	-
U3 - shared alpha	3	3.4	2	1.4	0.6	-
U3 - dose-specific alpha	3	3.4	2	1.4	0.6	-
Mean trial duration (days)						Days
U2 - shared alpha						932.8
U2 - dose-specific alpha						933.7
U3 - shared alpha						934.1
U3 - dose-specific alpha						932.7

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
MTD selection (%)						
U2 - shared alpha	0.7	0.6	2.3	8.4	88	0
U2 - dose-specific alpha	0.7	0.6	2.2	7.9	88.6	0
U3 - shared alpha	0.7	0.6	2.3	8.2	88.2	0
U3 - dose-specific alpha	0.7	0.6	2.5	7.5	88.7	0
OBD selection (%)						
U2 - shared alpha	31.3	23	16.7	16	12.9	0.1
U2 - dose-specific alpha	30.4	23.1	18.8	15.1	12.5	0.1
U3 - shared alpha	40.1	24.6	15.8	12.3	7.1	0.1
U3 - dose-specific alpha	39.5	26.6	15	12.5	6.3	0.1
Mean administrations						
U2 - shared alpha	6.1	5.4	5.5	5.8	7.3	-
U2 - dose-specific alpha	6	5.4	5.5	5.7	7.3	-
U3 - shared alpha	6	5.5	5.5	5.8	7.3	-
U3 - dose-specific alpha	6	5.5	5.5	5.8	7.3	-
Mean IPDE administrations						
U2 - shared alpha	1.3	3.1	1.9	2.6	2.8	-
U2 - dose-specific alpha	1.3	3.1	1.9	2.6	2.8	-
U3 - shared alpha	1.3	3.1	1.9	2.6	2.8	-
U3 - dose-specific alpha	1.3	3.1	1.9	2.6	2.8	-
Mean trial duration (days)						Days
U2 - shared alpha						1009
U2 - dose-specific alpha						1008.8
U3 - shared alpha						1008.4
U3 - dose-specific alpha						1007.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
MTD selection (%)						
U2 - shared alpha	14.7	37.4	39	8.5	0.2	0.2
U2 - dose-specific alpha	15.2	38.3	38.1	8	0.2	0.2
U3 - shared alpha	15.2	37.4	38.2	8.5	0.5	0.2
U3 - dose-specific alpha	15.1	37.9	38.5	8	0.3	0.2
OBD selection (%)						
U2 - shared alpha	30	49.3	17.7	2.1	0	0.9
U2 - dose-specific alpha	30.5	48.5	17.7	2.3	0	1
U3 - shared alpha	33.4	48.9	14.7	1.9	0.1	1
U3 - dose-specific alpha	34.1	49.6	13.8	1.5	0.1	0.9
Mean administrations						
U2 - shared alpha	10.1	9.7	7.2	2.4	0.4	-
U2 - dose-specific alpha	10.1	9.7	7.3	2.4	0.4	-
U3 - shared alpha	10.1	9.7	7.3	2.3	0.4	-
U3 - dose-specific alpha	10.1	9.6	7.3	2.4	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.5	3.4	1.6	0.6	0.1	-
U2 - dose-specific alpha	2.5	3.4	1.6	0.6	0.1	-
U3 - shared alpha	2.5	3.4	1.6	0.6	0.1	-
U3 - dose-specific alpha	2.5	3.4	1.6	0.6	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1191.4
U2 - dose-specific alpha						1191.8
U3 - shared alpha						1190.8
U3 - dose-specific alpha						1192.1

Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
MTD selection (%)						
U2 - shared alpha	0.7	0.1	3.3	21.5	74.4	0
U2 - dose-specific alpha	0.7	0.1	3.2	21.3	74.7	0
U3 - shared alpha	0.7	0.1	3.1	21.7	74.4	0
U3 - dose-specific alpha	0.7	0.1	3.2	20.8	75.2	0
OBD selection (%)						
U2 - shared alpha	9.1	8.3	9.6	13.6	58.3	1.1
U2 - dose-specific alpha	8.7	8.6	9.2	13.8	58.8	0.9
U3 - shared alpha	15.4	11.5	10.5	11.2	50.4	1
U3 - dose-specific alpha	15.4	11.8	9.9	11.4	50.6	0.9
Mean administrations						
U2 - shared alpha	3.7	3.9	4.6	5.5	12.2	-
U2 - dose-specific alpha	3.7	3.9	4.6	5.5	12.2	-
U3 - shared alpha	3.7	3.8	4.6	5.6	12.1	-
U3 - dose-specific alpha	3.7	3.9	4.5	5.5	12.2	-
Mean IPDE administrations						
U2 - shared alpha	0.3	2.7	1.5	2.5	3.4	-
U2 - dose-specific alpha	0.3	2.7	1.5	2.5	3.4	-
U3 - shared alpha	0.3	2.7	1.5	2.5	3.4	-
U3 - dose-specific alpha	0.3	2.7	1.5	2.5	3.4	-
Mean trial duration (days)						Days
U2 - shared alpha						1073.3
U2 - dose-specific alpha						1074.4
U3 - shared alpha						1071.8
U3 - dose-specific alpha						1073.1

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
MTD selection (%)						
U2 - shared alpha	19.6	39.6	34.7	5.7	0.2	0.2
U2 - dose-specific alpha	19.4	40	34.6	5.7	0.1	0.2
U3 - shared alpha	19.4	39.2	35.5	5.5	0.2	0.2
U3 - dose-specific alpha	18.8	40.2	34.4	6.3	0.1	0.2
OBD selection (%)						
U2 - shared alpha	16.1	50.6	17	1.7	0	14.6
U2 - dose-specific alpha	14.6	51.7	17.4	1.6	0	14.7
U3 - shared alpha	18.8	51	14.1	1.6	0	14.5
U3 - dose-specific alpha	18.1	50.8	14.9	1.8	0	14.4
Mean administrations						
U2 - shared alpha	7.6	10.2	7	2.2	0.4	-
U2 - dose-specific alpha	7.6	10.3	6.9	2.3	0.4	-
U3 - shared alpha	7.6	10.2	6.9	2.2	0.4	-
U3 - dose-specific alpha	7.6	10.2	7.1	2.2	0.4	-
Mean IPDE administrations						
U2 - shared alpha	1.9	4.4	2	0.8	0.1	-
U2 - dose-specific alpha	1.9	4.4	2	0.8	0.1	-
U3 - shared alpha	1.9	4.4	1.9	0.8	0.1	-
U3 - dose-specific alpha	1.9	4.4	2	0.8	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1006.2
U2 - dose-specific alpha						1006.9
U3 - shared alpha						1005.2
U3 - dose-specific alpha						1006.8

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
MTD selection (%)						
U2 - shared alpha	6.2	10.3	44.1	29	10.3	0.1
U2 - dose-specific alpha	6.1	10.6	44.1	28.2	10.9	0.1
U3 - shared alpha	6	10.5	46.8	26.2	10.4	0.1
U3 - dose-specific alpha	6	10.9	45.4	27.5	10.1	0.1
OBD selection (%)						
U2 - shared alpha	10.5	44	28.8	9.1	1.4	6.2
U2 - dose-specific alpha	11.2	42.1	28.9	10.2	1.4	6.2
U3 - shared alpha	16.4	46.5	22.6	7.6	0.8	6.1
U3 - dose-specific alpha	16.3	47.3	22.1	7.8	0.5	6
Mean administrations						
U2 - shared alpha	5.1	8.6	8.2	4.7	2.3	-
U2 - dose-specific alpha	5.2	8.6	8.2	4.6	2.2	-
U3 - shared alpha	5.1	8.6	8.3	4.6	2.2	-
U3 - dose-specific alpha	5.2	8.7	8.1	4.6	2.3	-
Mean IPDE administrations						
U2 - shared alpha	1	4.2	2.5	1.7	0.7	-
U2 - dose-specific alpha	1	4.3	2.5	1.7	0.7	-
U3 - shared alpha	1	4.3	2.5	1.7	0.7	-
U3 - dose-specific alpha	1	4.3	2.5	1.7	0.7	-
Mean trial duration (days)						Days
U2 - shared alpha						1025.2
U2 - dose-specific alpha						1025.4
U3 - shared alpha						1024.5
U3 - dose-specific alpha						1025.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
MTD selection (%)						
U2 - shared alpha	61.4	3.4	0.1	0	0	35.1
U2 - dose-specific alpha	62.1	3.4	0	0	0	34.5
U3 - shared alpha	62.2	3.2	0.1	0	0	34.5
U3 - dose-specific alpha	62.1	3.3	0	0	0	34.6
OBD selection (%)						
U2 - shared alpha	61.8	1.6	0	0	0	36.6
U2 - dose-specific alpha	61.4	1.8	0	0	0	36.8
U3 - shared alpha	62.1	1.3	0	0	0	36.6
U3 - dose-specific alpha	62	1.4	0	0	0	36.6
Mean administrations						
U2 - shared alpha	21.2	2.1	0.6	0	0	-
U2 - dose-specific alpha	21.1	2.1	0.6	0	0	-
U3 - shared alpha	21.1	2.1	0.6	0	0	-
U3 - dose-specific alpha	21.1	2.1	0.6	0	0	-
Mean IPDE administrations						
U2 - shared alpha	4.3	0.6	0.1	0	0	-
U2 - dose-specific alpha	4.3	0.6	0.1	0	0	-
U3 - shared alpha	4.3	0.6	0.1	0	0	-
U3 - dose-specific alpha	4.3	0.6	0.1	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1040.8
U2 - dose-specific alpha						1037.4
U3 - shared alpha						1040.2
U3 - dose-specific alpha						1039

Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
MTD selection (%)						
U2 - shared alpha	16.3	48.8	32.2	2.6	0	0.1
U2 - dose-specific alpha	16.4	50.2	30.3	3	0	0.1
U3 - shared alpha	16.2	49.6	31.8	2.3	0	0.1
U3 - dose-specific alpha	17.3	48.3	31.7	2.6	0	0.1
OBD selection (%)						
U2 - shared alpha	22.7	52.9	13	1.8	0	9.6
U2 - dose-specific alpha	23.5	52.4	12.4	2.1	0	9.6
U3 - shared alpha	25.8	52.2	10.8	1.4	0	9.8
U3 - dose-specific alpha	27.4	50.8	10.7	1.6	0	9.5
Mean administrations						
U2 - shared alpha	7.6	12.1	6.3	1.8	0.3	-
U2 - dose-specific alpha	7.6	12.2	6.4	1.7	0.3	-
U3 - shared alpha	7.6	12.1	6.5	1.7	0.3	-
U3 - dose-specific alpha	7.6	12.2	6.5	1.7	0.3	-
Mean IPDE administrations						
U2 - shared alpha	1.8	5.3	1.8	0.6	0.1	-
U2 - dose-specific alpha	1.8	5.2	1.9	0.6	0.1	-
U3 - shared alpha	1.8	5.2	1.9	0.6	0.1	-
U3 - dose-specific alpha	1.8	5.3	1.9	0.5	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1021.9
U2 - dose-specific alpha						1022.2
U3 - shared alpha						1021.8
U3 - dose-specific alpha						1024.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
MTD selection (%)						
U2 - shared alpha	79.8	6.9	0.5	0	0	12.8
U2 - dose-specific alpha	79.9	7	0.4	0	0	12.7
U3 - shared alpha	80.5	6.3	0.6	0	0	12.6
U3 - dose-specific alpha	80.1	6.8	0.5	0	0	12.6
OBD selection (%)						
U2 - shared alpha	7.2	3.9	0.4	0	0	88.5
U2 - dose-specific alpha	7.4	3.5	0.3	0	0	88.8
U3 - shared alpha	7.3	3.7	0.4	0	0	88.6
U3 - dose-specific alpha	7.4	3.7	0.4	0	0	88.5
Mean administrations						
U2 - shared alpha	11.8	2.5	0.6	0	0	-
U2 - dose-specific alpha	11.8	2.5	0.6	0	0	-
U3 - shared alpha	11.8	2.5	0.6	0	0	-
U3 - dose-specific alpha	11.8	2.5	0.6	0	0	-
Mean IPDE administrations						
U2 - shared alpha	3.3	1.3	0.2	0	0	-
U2 - dose-specific alpha	3.3	1.3	0.2	0	0	-
U3 - shared alpha	3.3	1.3	0.2	0	0	-
U3 - dose-specific alpha	3.3	1.3	0.2	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						547.3
U2 - dose-specific alpha						547.7
U3 - shared alpha						547.8
U3 - dose-specific alpha						547.6

Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
MTD selection (%)						
U2 - shared alpha	0	0.1	0.6	2.9	96.4	0
U2 - dose-specific alpha	0	0.1	0.6	2.6	96.7	0
U3 - shared alpha	0	0.1	0.5	3.3	96.1	0
U3 - dose-specific alpha	0	0	0.6	3.1	96.3	0
OBD selection (%)						
U2 - shared alpha	1.2	1.1	5.8	45.4	46.4	0.1
U2 - dose-specific alpha	1	1.3	5	47.2	45.4	0.1
U3 - shared alpha	2.8	2.2	7.5	47.3	40.1	0.1
U3 - dose-specific alpha	2.4	2	7.4	48.9	39.3	0
Mean administrations						
U2 - shared alpha	3.3	3.2	3.8	8.4	11.3	-
U2 - dose-specific alpha	3.3	3.2	3.8	8.4	11.3	-
U3 - shared alpha	3.3	3.2	3.8	8.4	11.3	-
U3 - dose-specific alpha	3.3	3.3	3.8	8.3	11.3	-
Mean IPDE administrations						
U2 - shared alpha	0.1	2.5	1.2	3.3	3.1	-
U2 - dose-specific alpha	0.1	2.5	1.2	3.3	3.1	-
U3 - shared alpha	0.1	2.5	1.2	3.3	3.1	-
U3 - dose-specific alpha	0.1	2.5	1.2	3.3	3.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1096.3
U2 - dose-specific alpha						1097.6
U3 - shared alpha						1096
U3 - dose-specific alpha						1096.5

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
MTD selection (%)						
U2 - shared alpha	11.1	29.2	48.3	10.9	0.3	0.2
U2 - dose-specific alpha	11.1	28.6	49.2	10.6	0.3	0.2
U3 - shared alpha	11.1	28.2	49.7	10.8	0	0.2
U3 - dose-specific alpha	10.9	29.4	48.6	10.7	0.2	0.2
OBD selection (%)						
U2 - shared alpha	31.9	26.8	33.5	5.3	0	2.5
U2 - dose-specific alpha	31.5	27.5	33.5	5.1	0	2.4
U3 - shared alpha	37.2	27.6	28.7	4.2	0	2.3
U3 - dose-specific alpha	37	28.2	28.3	3.9	0	2.6
Mean administrations						
U2 - shared alpha	9.1	8.1	9.2	2.8	0.5	-
U2 - dose-specific alpha	9.1	8	9.2	2.8	0.5	-
U3 - shared alpha	9.1	8	9.2	2.7	0.5	-
U3 - dose-specific alpha	9.1	8	9.2	2.8	0.5	-
Mean IPDE administrations						
U2 - shared alpha	2.4	3.6	2.5	0.8	0.1	-
U2 - dose-specific alpha	2.4	3.5	2.5	0.9	0.1	-
U3 - shared alpha	2.4	3.6	2.5	0.9	0.1	-
U3 - dose-specific alpha	2.4	3.5	2.5	0.9	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1113.3
U2 - dose-specific alpha						1113.8
U3 - shared alpha						1114.3
U3 - dose-specific alpha						1114.4

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
MTD selection (%)						
U2 - shared alpha	25	24.5	38.6	11.2	0.2	0.5
U2 - dose-specific alpha	25.1	24.7	38	11.5	0.2	0.5
U3 - shared alpha	24.8	24.6	38.4	11.5	0.2	0.5
U3 - dose-specific alpha	25	25.4	37.7	11.2	0.2	0.5
OBD selection (%)						
U2 - shared alpha	19.2	24	30.4	6	0	20.4
U2 - dose-specific alpha	19.7	25	28	6.8	0	20.5
U3 - shared alpha	23.1	26.4	24.4	6	0.1	20
U3 - dose-specific alpha	24	25	24.3	6.3	0	20.4
Mean administrations						
U2 - shared alpha	8.9	6.5	8	2.8	0.5	-
U2 - dose-specific alpha	9	6.5	7.9	2.9	0.5	-
U3 - shared alpha	9	6.6	7.9	2.9	0.5	-
U3 - dose-specific alpha	9	6.5	8	2.9	0.5	-
Mean IPDE administrations						
U2 - shared alpha	2.5	3.2	2.3	0.9	0.1	-
U2 - dose-specific alpha	2.5	3.2	2.3	0.9	0.1	-
U3 - shared alpha	2.5	3.2	2.3	0.9	0.1	-
U3 - dose-specific alpha	2.5	3.1	2.3	0.9	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						984.5
U2 - dose-specific alpha						983.3
U3 - shared alpha						985.3
U3 - dose-specific alpha						984.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
MTD selection (%)						
U2 - shared alpha	13.4	18.3	45.8	20.2	2.1	0.2
U2 - dose-specific alpha	13.5	17.2	47.7	19.1	2.3	0.2
U3 - shared alpha	13.2	18.5	46.1	19.6	2.4	0.2
U3 - dose-specific alpha	13.5	18.4	46.1	19.6	2.2	0.2
OBD selection (%)						
U2 - shared alpha	15.9	20.3	40.9	10.7	0.7	11.5
U2 - dose-specific alpha	14.8	21.6	41.5	9.5	1	11.6
U3 - shared alpha	20.2	22.2	36.7	8.9	0.7	11.3
U3 - dose-specific alpha	19.3	22.8	37.7	8.3	0.5	11.4
Mean administrations						
U2 - shared alpha	7.1	6.3	9.4	4.1	1.4	-
U2 - dose-specific alpha	7	6.4	9.4	4	1.4	-
U3 - shared alpha	7	6.3	9.5	4	1.4	-
U3 - dose-specific alpha	7	6.2	9.5	4	1.5	-
Mean IPDE administrations						
U2 - shared alpha	1.7	3.2	2.7	1.2	0.3	-
U2 - dose-specific alpha	1.7	3.1	2.7	1.2	0.4	-
U3 - shared alpha	1.8	3.1	2.7	1.2	0.3	-
U3 - dose-specific alpha	1.7	3.1	2.7	1.2	0.4	-
Mean trial duration (days)						Days
U2 - shared alpha						1050.3
U2 - dose-specific alpha						1051.8
U3 - shared alpha						1050.7
U3 - dose-specific alpha						1051.4

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
MTD selection (%)						
U2 - shared alpha	51.4	31.4	14.1	0.9	0	2.2
U2 - dose-specific alpha	50.9	31	14.8	1.1	0	2.2
U3 - shared alpha	50.5	32.1	14.1	1.1	0	2.2
U3 - dose-specific alpha	51	31.2	14.5	1.1	0	2.2
OBD selection (%)						
U2 - shared alpha	14.8	35.1	6.6	0.2	0	43.3
U2 - dose-specific alpha	13.9	35.5	6.5	0.5	0	43.6
U3 - shared alpha	15.5	35.5	5.5	0.3	0	43.2
U3 - dose-specific alpha	15.6	35.8	4.9	0.3	0	43.4
Mean administrations						
U2 - shared alpha	8.7	8.4	4.3	0.9	0.1	-
U2 - dose-specific alpha	8.8	8.3	4.3	0.9	0.1	-
U3 - shared alpha	8.8	8.4	4.2	0.9	0.1	-
U3 - dose-specific alpha	8.8	8.4	4.2	0.9	0.1	-
Mean IPDE administrations						
U2 - shared alpha	2.3	3.4	1.2	0.3	0	-
U2 - dose-specific alpha	2.3	3.4	1.2	0.3	0	-
U3 - shared alpha	2.3	3.4	1.2	0.3	0	-
U3 - dose-specific alpha	2.3	3.4	1.2	0.3	0	-
Mean trial duration (days)						Days
U2 - shared alpha						830.3
U2 - dose-specific alpha						830.2
U3 - shared alpha						829.2
U3 - dose-specific alpha						830.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
MTD selection (%)						
U2 - shared alpha	41.9	40.3	14.9	0.6	0	2.3
U2 - dose-specific alpha	42.3	39.6	14.8	1	0	2.3
U3 - shared alpha	42.1	39.6	15.2	0.7	0	2.4
U3 - dose-specific alpha	41.6	39.2	15.9	0.8	0	2.5
OBD selection (%)						
U2 - shared alpha	67.2	25.6	4.2	0.3	0	2.7
U2 - dose-specific alpha	69	23.6	4.4	0.3	0	2.7
U3 - shared alpha	69.8	23.7	3.4	0.3	0	2.8
U3 - dose-specific alpha	69.9	23.4	3.8	0.2	0	2.7
Mean administrations						
U2 - shared alpha	18	7.3	3.6	0.5	0	-
U2 - dose-specific alpha	18	7.3	3.5	0.6	0	-
U3 - shared alpha	18	7.3	3.6	0.5	0.1	-
U3 - dose-specific alpha	18.1	7.3	3.5	0.5	0	-
Mean IPDE administrations						
U2 - shared alpha	4.6	2.4	1	0.2	0	-
U2 - dose-specific alpha	4.6	2.4	1	0.2	0	-
U3 - shared alpha	4.6	2.4	1	0.2	0	-
U3 - dose-specific alpha	4.7	2.4	1	0.2	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1177.4
U2 - dose-specific alpha						1177.3
U3 - shared alpha						1178
U3 - dose-specific alpha						1177

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
MTD selection (%)						
U2 - shared alpha	2.1	1.9	20.5	46	29.5	0
U2 - dose-specific alpha	2.1	2	21.3	46.3	28.3	0
U3 - shared alpha	2.1	1.9	21.2	45.9	28.9	0
U3 - dose-specific alpha	2.1	1.9	20.9	46.1	29	0
OBD selection (%)						
U2 - shared alpha	19.2	13.8	15.1	37.1	13.1	1.7
U2 - dose-specific alpha	19.3	14	15.8	37.2	12	1.7
U3 - shared alpha	26.4	14.1	13.7	33.6	10.5	1.7
U3 - dose-specific alpha	26.7	15.4	12.7	33	10.5	1.7
Mean administrations						
U2 - shared alpha	5	4.5	5.8	8	6.4	-
U2 - dose-specific alpha	5	4.5	5.8	8	6.4	-
U3 - shared alpha	5	4.5	5.8	8	6.4	-
U3 - dose-specific alpha	5	4.5	5.7	8.2	6.3	-
Mean IPDE administrations						
U2 - shared alpha	0.9	2.7	1.8	2.8	1.6	-
U2 - dose-specific alpha	0.9	2.7	1.9	2.8	1.6	-
U3 - shared alpha	0.9	2.7	1.8	2.8	1.6	-
U3 - dose-specific alpha	0.9	2.7	1.8	2.9	1.5	-
Mean trial duration (days)						Days
U2 - shared alpha						1097.5
U2 - dose-specific alpha						1096
U3 - shared alpha						1097.5
U3 - dose-specific alpha						1097.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
MTD selection (%)						
U2 - shared alpha	1.5	1.2	7.9	38.8	50.6	0
U2 - dose-specific alpha	1.5	1.3	8.5	37.9	50.8	0
U3 - shared alpha	1.5	1.2	8.4	38.4	50.5	0
U3 - dose-specific alpha	1.5	1.2	8.4	38.7	50.2	0
OBD selection (%)						
U2 - shared alpha	7.2	34.9	24.8	24.9	8	0.2
U2 - dose-specific alpha	7.4	32.7	27.8	24.1	7.8	0.2
U3 - shared alpha	10.2	41.6	23.7	20.5	3.8	0.2
U3 - dose-specific alpha	9.7	41.8	24	20.5	3.8	0.2
Mean administrations						
U2 - shared alpha	4.1	5.7	6.4	7.7	6	-
U2 - dose-specific alpha	4.1	5.6	6.5	7.8	6	-
U3 - shared alpha	4.1	5.7	6.4	7.7	6	-
U3 - dose-specific alpha	4.1	5.7	6.4	7.7	6.1	-
Mean IPDE administrations						
U2 - shared alpha	0.5	2.8	1.6	2	1.5	-
U2 - dose-specific alpha	0.5	2.8	1.6	2	1.5	-
U3 - shared alpha	0.5	2.8	1.6	2	1.5	-
U3 - dose-specific alpha	0.5	2.8	1.6	2	1.5	-
Mean trial duration (days)						Days
U2 - shared alpha						1189.8
U2 - dose-specific alpha						1189.7
U3 - shared alpha						1188.8
U3 - dose-specific alpha						1189.2

Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
MTD selection (%)						
U2 - shared alpha	0.7	0.3	6.9	29.3	62.8	0
U2 - dose-specific alpha	0.7	0.4	6.7	28.8	63.4	0
U3 - shared alpha	0.7	0.3	6.5	29.4	63.1	0
U3 - dose-specific alpha	0.7	0.3	6.5	29	63.5	0
OBD selection (%)						
U2 - shared alpha	8.7	13.4	34.7	27.2	15.1	0.9
U2 - dose-specific alpha	8.5	13.8	35.6	26.7	14.5	0.9
U3 - shared alpha	14.6	17.8	34.6	23.2	9	0.8
U3 - dose-specific alpha	14.5	16.6	34.7	24.1	9.2	0.9
Mean administrations						
U2 - shared alpha	3.9	4.3	7.7	7.1	6.9	-
U2 - dose-specific alpha	4	4.3	7.7	7.1	6.9	-
U3 - shared alpha	4	4.3	7.7	7	6.9	-
U3 - dose-specific alpha	4	4.3	7.6	7	6.9	-
Mean IPDE administrations						
U2 - shared alpha	0.4	2.8	2.5	2.7	2.3	-
U2 - dose-specific alpha	0.4	2.8	2.4	2.7	2.3	-
U3 - shared alpha	0.4	2.8	2.5	2.7	2.3	-
U3 - dose-specific alpha	0.4	2.8	2.4	2.7	2.3	-
Mean trial duration (days)						Days
U2 - shared alpha						1054.9
U2 - dose-specific alpha						1055.6
U3 - shared alpha						1055
U3 - dose-specific alpha						1055.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
MTD selection (%)						
U2 - shared alpha	0	0.5	8.1	36	55.4	0
U2 - dose-specific alpha	0	0.4	7.2	36.9	55.5	0
U3 - shared alpha	0	0.5	7.4	37.1	55	0
U3 - dose-specific alpha	0	0.4	7.3	36.3	56	0
OBD selection (%)						
U2 - shared alpha	8.6	5.1	11.6	41.1	33.3	0.3
U2 - dose-specific alpha	8.4	4.8	11.8	40.9	33.8	0.3
U3 - shared alpha	12.3	6.9	13.6	40	27	0.2
U3 - dose-specific alpha	13.1	6	13.2	40.1	27.3	0.3
Mean administrations						
U2 - shared alpha	3.5	3.6	5.2	8.1	9.6	-
U2 - dose-specific alpha	3.5	3.6	5.2	7.9	9.7	-
U3 - shared alpha	3.5	3.6	5.2	8.1	9.6	-
U3 - dose-specific alpha	3.5	3.6	5.2	8	9.7	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.6	1.6	3.1	2.4	-
U2 - dose-specific alpha	0.2	2.6	1.6	3	2.4	-
U3 - shared alpha	0.2	2.6	1.6	3.1	2.4	-
U3 - dose-specific alpha	0.2	2.6	1.6	3	2.4	-
Mean trial duration (days)						Days
U2 - shared alpha						1107.4
U2 - dose-specific alpha						1106.6
U3 - shared alpha						1106.9
U3 - dose-specific alpha						1106.4

Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
MTD selection (%)						
U2 - shared alpha	2.4	8.8	40.7	30	18.1	0
U2 - dose-specific alpha	2.5	8.6	41.7	29.3	17.9	0
U3 - shared alpha	2.4	9	40.9	28.9	18.8	0
U3 - dose-specific alpha	2.5	8.8	40.6	29.8	18.3	0
OBD selection (%)						
U2 - shared alpha	11.9	11.7	47	19.8	5.4	4.2
U2 - dose-specific alpha	12.7	11.3	47.7	18.4	5.7	4.2
U3 - shared alpha	18.2	13.3	44.3	16.1	3.6	4.5
U3 - dose-specific alpha	18.2	13.1	44.8	16.7	2.9	4.3
Mean administrations						
U2 - shared alpha	5	5.5	9.9	5.4	3.4	-
U2 - dose-specific alpha	5	5.5	9.8	5.5	3.5	-
U3 - shared alpha	5	5.5	9.8	5.4	3.5	-
U3 - dose-specific alpha	5	5.5	9.8	5.3	3.5	-
Mean IPDE administrations						
U2 - shared alpha	0.9	3.2	3.1	1.9	1.1	-
U2 - dose-specific alpha	0.9	3.2	3.1	1.9	1.1	-
U3 - shared alpha	0.9	3.2	3.1	1.9	1.1	-
U3 - dose-specific alpha	0.9	3.2	3.1	1.9	1.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1049.8
U2 - dose-specific alpha						1049
U3 - shared alpha						1050.6
U3 - dose-specific alpha						1049.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
MTD selection (%)						
U2 - shared alpha	1.2	6.1	41.3	31.4	20	0
U2 - dose-specific alpha	1.4	5.1	42.5	31.7	19.3	0
U3 - shared alpha	1.3	5.9	40.8	33.1	18.9	0
U3 - dose-specific alpha	1.3	6.1	41	32	19.6	0
OBD selection (%)						
U2 - shared alpha	11.6	40.2	31.4	12.3	4.3	0.2
U2 - dose-specific alpha	12.7	37.5	32.3	12.8	4.5	0.2
U3 - shared alpha	16.1	44	27.2	9.9	2.6	0.2
U3 - dose-specific alpha	17	43	27.5	9.8	2.5	0.2
Mean administrations						
U2 - shared alpha	4.6	8.3	8.8	5	3.3	-
U2 - dose-specific alpha	4.6	8.2	9	5	3.2	-
U3 - shared alpha	4.6	8.2	8.9	5	3.2	-
U3 - dose-specific alpha	4.7	8.2	8.9	5	3.2	-
Mean IPDE administrations						
U2 - shared alpha	0.7	3.7	2.2	1.5	0.9	-
U2 - dose-specific alpha	0.6	3.6	2.2	1.5	0.8	-
U3 - shared alpha	0.6	3.6	2.2	1.5	0.8	-
U3 - dose-specific alpha	0.7	3.6	2.2	1.5	0.9	-
Mean trial duration (days)						Days
U2 - shared alpha						1166.3
U2 - dose-specific alpha						1166.3
U3 - shared alpha						1166.4
U3 - dose-specific alpha						1166.2

Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
MTD selection (%)						
U2 - shared alpha	25.7	30.1	24.7	13.6	5.5	0.4
U2 - dose-specific alpha	25.5	30.3	25.2	13	5.6	0.4
U3 - shared alpha	26	30.4	24.1	13.3	5.8	0.4
U3 - dose-specific alpha	25.2	30.5	25.1	13.2	5.6	0.4
OBD selection (%)						
U2 - shared alpha	14.6	36.6	16.1	7	1.8	23.9
U2 - dose-specific alpha	14.9	36.5	16.1	6.4	1.8	24.3
U3 - shared alpha	19.1	37.8	12.6	5.9	1	23.6
U3 - dose-specific alpha	18.9	37.2	13.2	5.7	1.1	23.9
Mean administrations						
U2 - shared alpha	7.8	8	5.7	2.9	1.4	-
U2 - dose-specific alpha	7.7	8.1	5.7	2.8	1.4	-
U3 - shared alpha	7.6	8.2	5.7	2.8	1.4	-
U3 - dose-specific alpha	7.7	8.1	5.7	2.9	1.4	-
Mean IPDE administrations						
U2 - shared alpha	2	3.8	1.8	1.1	0.4	-
U2 - dose-specific alpha	1.9	3.9	1.8	1.1	0.4	-
U3 - shared alpha	1.9	3.9	1.8	1	0.4	-
U3 - dose-specific alpha	1.9	3.8	1.8	1.1	0.4	-
Mean trial duration (days)						Days
U2 - shared alpha						914.5
U2 - dose-specific alpha						915.4
U3 - shared alpha						918.1
U3 - dose-specific alpha						915.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = two-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
MTD selection (%)						
U2 - shared alpha	19.8	39.8	28.1	9.4	2.8	0.1
U2 - dose-specific alpha	20	40.4	27.2	9.4	2.9	0.1
U3 - shared alpha	19.5	39.5	28.6	9.2	3.1	0.1
U3 - dose-specific alpha	19.8	40.1	27.6	9.9	2.5	0.1
OBD selection (%)						
U2 - shared alpha	20.4	45.6	14.7	2.7	0.5	16.1
U2 - dose-specific alpha	21.3	45.2	13.2	3.2	0.7	16.4
U3 - shared alpha	25.5	44.6	11.2	2	0.4	16.3
U3 - dose-specific alpha	26.6	44.3	10	2.3	0.3	16.5
Mean administrations						
U2 - shared alpha	7.3	10.3	6.2	2.4	0.9	-
U2 - dose-specific alpha	7.2	10.4	6.2	2.3	1	-
U3 - shared alpha	7.2	10.3	6.2	2.4	0.9	-
U3 - dose-specific alpha	7.2	10.4	6.1	2.4	1	-
Mean IPDE administrations						
U2 - shared alpha	1.7	4.8	1.9	1	0.3	-
U2 - dose-specific alpha	1.7	4.8	1.9	0.9	0.3	-
U3 - shared alpha	1.7	4.8	1.9	0.9	0.3	-
U3 - dose-specific alpha	1.6	4.9	1.9	0.9	0.3	-
Mean trial duration (days)						Days
U2 - shared alpha						955.2
U2 - dose-specific alpha						955.3
U3 - shared alpha						954.7
U3 - dose-specific alpha						955.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 1						
DLT rate	0.30	0.35	0.40	0.45	0.50	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
MTD selection (%)						
U2 - shared alpha	55.4	22.3	9	4.3	1	8
U2 - dose-specific alpha	55.2	23.1	8.5	4.5	0.9	7.8
U3 - shared alpha	55.9	22.4	8.4	4.1	1.3	7.9
U3 - dose-specific alpha	55.5	22.4	8.8	4.2	1	8.1
OBD selection (%)						
U2 - shared alpha	16.4	11.5	2.3	0.6	0.2	69
U2 - dose-specific alpha	16	12.2	2.5	0.7	0.2	68.4
U3 - shared alpha	16.7	11.4	1.8	0.4	0.1	69.6
U3 - dose-specific alpha	17.2	11	2	0.4	0.1	69.3
Mean administrations						
U2 - shared alpha	10.7	5.7	2.5	1.4	0.4	-
U2 - dose-specific alpha	10.7	5.7	2.5	1.4	0.4	-
U3 - shared alpha	10.8	5.6	2.5	1.4	0.4	-
U3 - dose-specific alpha	10.8	5.7	2.5	1.4	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.8	2.9	0.8	0.6	0.2	-
U2 - dose-specific alpha	2.8	2.9	0.8	0.6	0.2	-
U3 - shared alpha	2.9	2.9	0.8	0.6	0.2	-
U3 - dose-specific alpha	2.9	2.9	0.8	0.6	0.2	-
Mean trial duration (days)						Days
U2 - shared alpha						732.2
U2 - dose-specific alpha						730.9
U3 - shared alpha						731.2
U3 - dose-specific alpha						733.4

Scenario 2						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.20	0.25	0.50	0.50	0.50	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
MTD selection (%)						
U2 - shared alpha	4.9	25.9	40.3	23	5.6	0.3
U2 - dose-specific alpha	5.1	27	38.2	23.6	5.8	0.3
U3 - shared alpha	5.2	28	39.9	21.6	5	0.3
U3 - dose-specific alpha	4.8	28.6	40.2	20.9	5.2	0.3
OBD selection (%)						
U2 - shared alpha	23	17.3	39.6	15.4	2.1	2.6
U2 - dose-specific alpha	24.4	17.7	38.8	14.6	2.1	2.4
U3 - shared alpha	28.1	19	35.2	13.1	1.7	2.9
U3 - dose-specific alpha	28.7	18.4	35.7	12.8	1.7	2.7
Mean administrations						
U2 - shared alpha	9	6.9	8.9	3.9	0.9	-
U2 - dose-specific alpha	9	6.9	8.9	3.9	0.9	-
U3 - shared alpha	9.3	7.1	8.8	3.6	0.7	-
U3 - dose-specific alpha	9.4	7.1	8.7	3.6	0.8	-
Mean IPDE administrations						
U2 - shared alpha	2.2	3.2	2.4	1.1	0.3	-
U2 - dose-specific alpha	2.2	3.2	2.4	1.1	0.3	-
U3 - shared alpha	2.3	3.3	2.4	1.1	0.2	-
U3 - dose-specific alpha	2.4	3.3	2.4	1.1	0.2	-
Mean trial duration (days)						Days
U2 - shared alpha						1125.4
U2 - dose-specific alpha						1125.9
U3 - shared alpha						1120.4
U3 - dose-specific alpha						1121.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 3						
DLT rate	0.15	0.20	0.30	0.35	0.45	
Efficacy rate	0.10	0.40	0.40	0.40	0.40	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
MTD selection (%)						
U2 - shared alpha	8.5	24.1	35.9	23.8	7.4	0.3
U2 - dose-specific alpha	8.5	24.3	35.9	23.4	7.6	0.3
U3 - shared alpha	8.5	27.1	35.8	22.4	5.9	0.3
U3 - dose-specific alpha	8.4	27.2	35.7	22.1	6.3	0.3
OBD selection (%)						
U2 - shared alpha	7.9	44.5	24.9	13.2	2.4	7.1
U2 - dose-specific alpha	8.4	44.2	25.9	11.9	2.4	7.2
U3 - shared alpha	9.6	49.1	23.1	9.2	1.7	7.3
U3 - dose-specific alpha	9.9	50	21.5	9.7	1.7	7.2
Mean administrations						
U2 - shared alpha	6.1	11.1	6.8	3.8	1.1	-
U2 - dose-specific alpha	6.1	10.9	6.8	3.9	1.2	-
U3 - shared alpha	6.3	11.8	6.5	3.4	0.9	-
U3 - dose-specific alpha	6.3	11.8	6.5	3.4	1	-
Mean IPDE administrations						
U2 - shared alpha	1.1	4.6	2	1.2	0.3	-
U2 - dose-specific alpha	1.1	4.5	2	1.3	0.4	-
U3 - shared alpha	1.1	4.8	1.9	1.1	0.3	-
U3 - dose-specific alpha	1.1	4.8	1.9	1.1	0.3	-
Mean trial duration (days)						Days
U2 - shared alpha						1085
U2 - dose-specific alpha						1085.5
U3 - shared alpha						1081.2
U3 - dose-specific alpha						1081.9

Scenario 4						
DLT rate	0.05	0.10	0.18	0.30	0.40	
Efficacy rate	0.05	0.10	0.25	0.50	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
MTD selection (%)						
U2 - shared alpha	0.6	1.7	14.9	54.2	28.6	0
U2 - dose-specific alpha	0.6	1.8	14.9	53.2	29.5	0
U3 - shared alpha	0.6	2.2	14.8	56	26.4	0
U3 - dose-specific alpha	0.6	1.8	15.5	55.8	26.3	0
OBD selection (%)						
U2 - shared alpha	5.7	5.2	16.9	53.3	17.6	1.3
U2 - dose-specific alpha	4.8	5.9	16.8	53	18.2	1.3
U3 - shared alpha	8.6	7.7	16.4	51.3	14.9	1.1
U3 - dose-specific alpha	9.7	7.6	16.2	51.1	14.3	1.1
Mean administrations						
U2 - shared alpha	4.1	4.5	6.7	10.6	3.9	-
U2 - dose-specific alpha	4.1	4.5	6.7	10.5	4	-
U3 - shared alpha	4.3	4.6	6.7	10.7	3.6	-
U3 - dose-specific alpha	4.2	4.6	6.8	10.7	3.5	-
Mean IPDE administrations						
U2 - shared alpha	0.5	3	1.9	3.3	1	-
U2 - dose-specific alpha	0.5	3	1.9	3.3	1.1	-
U3 - shared alpha	0.5	3	1.9	3.4	0.9	-
U3 - dose-specific alpha	0.5	3	2	3.4	0.9	-
Mean trial duration (days)						Days
U2 - shared alpha						1107.3
U2 - dose-specific alpha						1107.9
U3 - shared alpha						1104.7
U3 - dose-specific alpha						1104.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.30	
Efficacy rate	0.05	0.30	0.30	0.30	0.30	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
MTD selection (%)						
U2 - shared alpha	1.4	2.7	13.3	28.2	54.4	0
U2 - dose-specific alpha	1.4	2.5	12.5	29.1	54.5	0
U3 - shared alpha	1.5	3.4	16.2	29.7	49.2	0
U3 - dose-specific alpha	1.4	3.7	15.3	30.8	48.8	0
OBD selection (%)						
U2 - shared alpha	5.4	37.6	25.7	19.5	10.4	1.4
U2 - dose-specific alpha	5.8	35.9	26.1	19.9	10.9	1.4
U3 - shared alpha	9	42.6	23.2	17.5	6.2	1.5
U3 - dose-specific alpha	8.4	43.5	23.6	17.3	5.8	1.4
Mean administrations						
U2 - shared alpha	3.8	7.2	7	7.1	4.6	-
U2 - dose-specific alpha	3.8	7.1	7	7.2	4.7	-
U3 - shared alpha	3.9	7.9	7.2	6.8	4	-
U3 - dose-specific alpha	3.9	7.8	7.1	6.9	4.1	-
Mean IPDE administrations						
U2 - shared alpha	0.3	3.8	2.3	2.7	1.6	-
U2 - dose-specific alpha	0.3	3.8	2.2	2.7	1.7	-
U3 - shared alpha	0.3	4.1	2.3	2.6	1.4	-
U3 - dose-specific alpha	0.3	4.1	2.3	2.6	1.4	-
Mean trial duration (days)						Days
U2 - shared alpha						1052.3
U2 - dose-specific alpha						1052.1
U3 - shared alpha						1049.9
U3 - dose-specific alpha						1050.1

Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.30	0.32	0.35	0.38	0.40	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
MTD selection (%)						
U2 - shared alpha	45.2	29.1	11.4	4.4	0.8	9.1
U2 - dose-specific alpha	44.8	29	11.1	4.7	0.7	9.7
U3 - shared alpha	45	30.7	9.5	4.5	0.7	9.6
U3 - dose-specific alpha	44.8	30.7	9.9	4.4	0.5	9.7
OBD selection (%)						
U2 - shared alpha	61.9	17.6	5.1	1.9	0.5	13
U2 - dose-specific alpha	61	18.6	5.1	1.8	0.5	13
U3 - shared alpha	62.7	17.8	4.2	1.5	0.4	13.4
U3 - dose-specific alpha	63.7	17.6	3.7	1.4	0.3	13.3
Mean administrations						
U2 - shared alpha	17.1	6.3	2.7	1.3	0.3	-
U2 - dose-specific alpha	17.1	6.3	2.7	1.3	0.3	-
U3 - shared alpha	17.6	6.2	2.4	1.2	0.3	-
U3 - dose-specific alpha	17.7	6.2	2.4	1.2	0.3	-
Mean IPDE administrations						
U2 - shared alpha	4.3	2.5	0.8	0.4	0.1	-
U2 - dose-specific alpha	4.3	2.5	0.8	0.4	0.1	-
U3 - shared alpha	4.5	2.5	0.7	0.4	0.1	-
U3 - dose-specific alpha	4.5	2.5	0.7	0.4	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1081.9
U2 - dose-specific alpha						1081.6
U3 - shared alpha						1078.3
U3 - dose-specific alpha						1079.7

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 7						
DLT rate	0.16	0.20	0.26	0.31	0.36	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
MTD selection (%)						
U2 - shared alpha	6.7	37.7	29.7	18.8	6.7	0.4
U2 - dose-specific alpha	6.3	37.3	30.3	19.4	6.3	0.4
U3 - shared alpha	6.6	43.2	29	15.3	5.5	0.4
U3 - dose-specific alpha	6.4	43.3	28.7	16	5.2	0.4
OBD selection (%)						
U2 - shared alpha	11.8	46.8	22.7	12.5	3.5	2.7
U2 - dose-specific alpha	11.8	46.5	23.6	12	3.3	2.8
U3 - shared alpha	13.5	50.5	20.7	10.2	2.3	2.8
U3 - dose-specific alpha	14.3	49.8	21.2	9.8	2.1	2.8
Mean administrations						
U2 - shared alpha	7.4	12.5	5.8	3	0.8	-
U2 - dose-specific alpha	7.4	12.5	6	2.9	0.8	-
U3 - shared alpha	7.7	13.2	5.4	2.6	0.7	-
U3 - dose-specific alpha	7.7	13.1	5.5	2.5	0.7	-
Mean IPDE administrations						
U2 - shared alpha	1.4	4.6	1.6	0.9	0.2	-
U2 - dose-specific alpha	1.4	4.6	1.6	0.9	0.2	-
U3 - shared alpha	1.5	4.8	1.4	0.8	0.2	-
U3 - dose-specific alpha	1.6	4.8	1.4	0.8	0.2	-
Mean trial duration (days)						Days
U2 - shared alpha						1151.9
U2 - dose-specific alpha						1151.5
U3 - shared alpha						1150.1
U3 - dose-specific alpha						1149.8

Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.10	0.15	0.20	0.30	0.40	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
MTD selection (%)						
U2 - shared alpha	56.5	20	7.7	2.3	0.6	12.9
U2 - dose-specific alpha	56.7	19.1	8.6	2.2	0.7	12.7
U3 - shared alpha	57.6	19.7	7.7	2.2	0.7	12.1
U3 - dose-specific alpha	57.7	19.6	8.2	1.7	0.7	12.1
OBD selection (%)						
U2 - shared alpha	28.7	12.4	3.5	1.5	0.3	53.6
U2 - dose-specific alpha	28.4	13.5	3.6	1.2	0.3	53
U3 - shared alpha	28.2	14	2.4	1.3	0.2	53.9
U3 - dose-specific alpha	28	14.2	2.4	1.1	0.2	54.1
Mean administrations						
U2 - shared alpha	12.6	6	2.3	1.2	0.3	-
U2 - dose-specific alpha	12.7	5.9	2.3	1.2	0.3	-
U3 - shared alpha	12.9	5.8	2.2	1.1	0.3	-
U3 - dose-specific alpha	12.8	5.9	2.2	1.1	0.3	-
Mean IPDE administrations						
U2 - shared alpha	3.3	2.8	0.7	0.4	0.1	-
U2 - dose-specific alpha	3.4	2.8	0.7	0.4	0.1	-
U3 - shared alpha	3.5	2.8	0.7	0.4	0.1	-
U3 - dose-specific alpha	3.4	2.8	0.7	0.4	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						818.9
U2 - dose-specific alpha						821.9
U3 - shared alpha						820
U3 - dose-specific alpha						816.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.20	0.25	0.50	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
MTD selection (%)						
U2 - shared alpha	8.6	9.3	17	25.8	39	0.3
U2 - dose-specific alpha	8.7	9.6	16.5	26.4	38.5	0.3
U3 - shared alpha	8.7	10	17.1	25.4	38.5	0.3
U3 - dose-specific alpha	8.7	9.7	16.6	26.6	38.1	0.3
OBD selection (%)						
U2 - shared alpha	10.4	11	21.2	17	27.6	12.8
U2 - dose-specific alpha	11.4	10.9	20.8	16	27.5	13.4
U3 - shared alpha	15.4	13.2	19.8	15.1	23.3	13.2
U3 - dose-specific alpha	15.2	13.1	20.5	14.6	23.7	12.9
Mean administrations						
U2 - shared alpha	6.2	5.6	6	5.4	5.2	-
U2 - dose-specific alpha	6.2	5.6	6	5.4	5.2	-
U3 - shared alpha	6.4	5.7	5.9	5.4	4.8	-
U3 - dose-specific alpha	6.4	5.7	6	5.4	4.9	-
Mean IPDE administrations						
U2 - shared alpha	1.3	3.3	1.9	2.1	1.6	-
U2 - dose-specific alpha	1.3	3.3	2	2.1	1.6	-
U3 - shared alpha	1.4	3.3	1.9	2.1	1.5	-
U3 - dose-specific alpha	1.3	3.3	2	2.1	1.5	-
Mean trial duration (days)						Days
U2 - shared alpha						999.7
U2 - dose-specific alpha						999.4
U3 - shared alpha						997
U3 - dose-specific alpha						997.6

Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.40	0.42	0.45	0.50	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
MTD selection (%)						
U2 - shared alpha	34.8	25.1	21.6	12.6	3.5	2.4
U2 - dose-specific alpha	34.8	25.3	21.6	12.8	3.2	2.3
U3 - shared alpha	35	26.4	22.1	11.4	2.9	2.2
U3 - dose-specific alpha	34.9	26.7	22	11.4	2.6	2.4
OBD selection (%)						
U2 - shared alpha	4.7	36.3	15.9	6.5	1.8	34.8
U2 - dose-specific alpha	4.6	35.2	17.1	6.6	1.7	34.8
U3 - shared alpha	5.2	38	15.1	5.5	1.1	35.1
U3 - dose-specific alpha	5.4	38.7	14.9	5.2	1	34.8
Mean administrations						
U2 - shared alpha	7.5	9.7	4.8	2.2	0.7	-
U2 - dose-specific alpha	7.5	9.7	4.8	2.3	0.6	-
U3 - shared alpha	7.6	10.1	4.6	2.1	0.6	-
U3 - dose-specific alpha	7.6	10.2	4.6	2	0.5	-
Mean IPDE administrations						
U2 - shared alpha	1.5	4.1	1.3	0.7	0.2	-
U2 - dose-specific alpha	1.5	4.1	1.3	0.7	0.2	-
U3 - shared alpha	1.5	4.3	1.2	0.7	0.2	-
U3 - dose-specific alpha	1.5	4.3	1.2	0.7	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						944.9
U2 - dose-specific alpha						943.9
U3 - shared alpha						944.8
U3 - dose-specific alpha						944.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
MTD selection (%)						
U2 - shared alpha	32.6	45.9	16.7	2.6	0	2.2
U2 - dose-specific alpha	32.6	45	17.3	2.9	0	2.2
U3 - shared alpha	31.4	46.4	17	2.9	0	2.3
U3 - dose-specific alpha	31.7	45.4	17.3	3	0	2.6
OBD selection (%)						
U2 - shared alpha	57.3	23.4	4.4	0.6	0	14.3
U2 - dose-specific alpha	56.8	23.2	5.2	0.7	0	14.1
U3 - shared alpha	59.9	21.1	4.5	0.6	0	13.9
U3 - dose-specific alpha	60.6	21.2	3.9	0.6	0	13.7
Mean administrations						
U2 - shared alpha	15.4	7.8	3.4	1.4	0.2	-
U2 - dose-specific alpha	15.4	7.8	3.4	1.4	0.2	-
U3 - shared alpha	15.9	7.6	3.2	1.3	0.2	-
U3 - dose-specific alpha	15.9	7.6	3.2	1.3	0.2	-
Mean IPDE administrations						
U2 - shared alpha	4.4	3.5	1	0.5	0.1	-
U2 - dose-specific alpha	4.4	3.5	1	0.5	0.1	-
U3 - shared alpha	4.6	3.5	0.9	0.5	0.1	-
U3 - dose-specific alpha	4.6	3.5	0.9	0.5	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1031.3
U2 - dose-specific alpha						1033.8
U3 - shared alpha						1032.1
U3 - dose-specific alpha						1033.3

Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.10	0.12	0.40	0.60	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
MTD selection (%)						
U2 - shared alpha	0.9	2.1	14	49	34	0
U2 - dose-specific alpha	0.9	2.1	13.8	49.5	33.7	0
U3 - shared alpha	0.9	2.1	13.7	52	31.3	0
U3 - dose-specific alpha	0.9	2.3	13.9	51	31.9	0
OBD selection (%)						
U2 - shared alpha	10.7	10.7	8.3	43.9	24.3	2.1
U2 - dose-specific alpha	10.4	11.4	8.2	44.3	23.5	2.2
U3 - shared alpha	14.5	13.6	7.1	42.4	20.4	2
U3 - dose-specific alpha	14.8	13.7	6.9	41.1	21.5	2
Mean administrations						
U2 - shared alpha	4.4	5	5.4	9.5	5.5	-
U2 - dose-specific alpha	4.4	5	5.5	9.4	5.5	-
U3 - shared alpha	4.6	5.1	5.4	9.6	5.2	-
U3 - dose-specific alpha	4.6	5.1	5.4	9.5	5.2	-
Mean IPDE administrations						
U2 - shared alpha	0.6	3.2	1.6	3.3	1.4	-
U2 - dose-specific alpha	0.6	3.2	1.6	3.3	1.4	-
U3 - shared alpha	0.7	3.2	1.6	3.3	1.3	-
U3 - dose-specific alpha	0.7	3.2	1.6	3.3	1.3	-
Mean trial duration (days)						Days
U2 - shared alpha						1084
U2 - dose-specific alpha						1084.3
U3 - shared alpha						1079.6
U3 - dose-specific alpha						1081

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 13						
DLT rate	0.21	0.29	0.30	0.39	0.46	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
MTD selection (%)						
U2 - shared alpha	32.1	24	21.4	16.7	4.1	1.7
U2 - dose-specific alpha	31.3	24.5	22.4	16.1	4	1.7
U3 - shared alpha	32.1	24.9	21.7	16.3	3.2	1.8
U3 - dose-specific alpha	32.5	24.3	22.5	15.4	3.6	1.7
OBD selection (%)						
U2 - shared alpha	6.3	36.8	17.2	6.7	1.7	31.3
U2 - dose-specific alpha	6	37	17.9	6.4	1.8	30.9
U3 - shared alpha	8.5	37.6	15.9	5.1	1.3	31.6
U3 - dose-specific alpha	7.8	38.4	15.1	5.4	1.2	32.1
Mean administrations						
U2 - shared alpha	7.6	9.1	5	2.9	0.9	-
U2 - dose-specific alpha	7.6	9.1	5.1	2.9	0.9	-
U3 - shared alpha	7.7	9.4	4.9	2.7	0.8	-
U3 - dose-specific alpha	7.7	9.3	4.9	2.7	0.8	-
Mean IPDE administrations						
U2 - shared alpha	1.5	4.2	1.5	1	0.3	-
U2 - dose-specific alpha	1.5	4.2	1.5	1	0.3	-
U3 - shared alpha	1.6	4.3	1.4	1	0.3	-
U3 - dose-specific alpha	1.6	4.3	1.4	1	0.3	-
Mean trial duration (days)						Days
U2 - shared alpha						932.7
U2 - dose-specific alpha						934.4
U3 - shared alpha						930
U3 - dose-specific alpha						930.2

Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.30	0.40	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
MTD selection (%)						
U2 - shared alpha	2.3	6.6	20.5	36.6	34	0
U2 - dose-specific alpha	2.5	6.7	21.4	34.9	34.5	0
U3 - shared alpha	2.6	6.9	20.7	37.7	32.1	0
U3 - dose-specific alpha	2.5	6.8	20.7	37.3	32.7	0
OBD selection (%)						
U2 - shared alpha	14.9	15.1	14.4	34.2	16	5.4
U2 - dose-specific alpha	15.2	14.9	14.4	34.2	15.8	5.5
U3 - shared alpha	21.3	18.2	13.2	30.6	11.2	5.5
U3 - dose-specific alpha	20.9	17.3	14.5	30.2	11.5	5.6
Mean administrations						
U2 - shared alpha	5.7	5.9	5.7	7.8	4.3	-
U2 - dose-specific alpha	5.7	5.9	5.8	7.7	4.3	-
U3 - shared alpha	6	6.1	5.6	7.8	4	-
U3 - dose-specific alpha	6	6.1	5.6	7.7	4	-
Mean IPDE administrations						
U2 - shared alpha	1.1	3.5	1.8	3	1.4	-
U2 - dose-specific alpha	1.2	3.5	1.9	2.9	1.4	-
U3 - shared alpha	1.3	3.6	1.8	2.9	1.3	-
U3 - dose-specific alpha	1.3	3.6	1.8	2.9	1.3	-
Mean trial duration (days)						Days
U2 - shared alpha						1028.1
U2 - dose-specific alpha						1027.4
U3 - shared alpha						1024.9
U3 - dose-specific alpha						1024.5

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 15						
DLT rate	0.13	0.20	0.32	0.39	0.46	
Efficacy rate	0.10	0.10	0.10	0.10	0.10	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
MTD selection (%)						
U2 - shared alpha	6.3	24.6	34.6	24.5	9.8	0.2
U2 - dose-specific alpha	6.7	24.4	33.1	25.4	10.2	0.2
U3 - shared alpha	6.2	24.3	33.5	25.5	10.3	0.2
U3 - dose-specific alpha	6	24.6	34	25.2	10	0.2
OBD selection (%)						
U2 - shared alpha	41.9	27.2	13.1	6.5	0.4	10.9
U2 - dose-specific alpha	41.1	28	12.2	6.7	0.4	11.6
U3 - shared alpha	46.1	28.2	10	3.3	0.5	11.9
U3 - dose-specific alpha	46.3	27.7	10.4	3.7	0.3	11.6
Mean administrations						
U2 - shared alpha	9.4	8	5.7	4.1	1.6	-
U2 - dose-specific alpha	9.4	8	5.8	4.1	1.5	-
U3 - shared alpha	9.9	8	5.5	4	1.5	-
U3 - dose-specific alpha	9.9	8.1	5.5	3.9	1.5	-
Mean IPDE administrations						
U2 - shared alpha	2.6	4.2	1.9	1.7	0.5	-
U2 - dose-specific alpha	2.6	4.2	1.9	1.7	0.5	-
U3 - shared alpha	2.9	4.2	1.8	1.7	0.5	-
U3 - dose-specific alpha	2.9	4.2	1.8	1.7	0.5	-
Mean trial duration (days)						Days
U2 - shared alpha						982.4
U2 - dose-specific alpha						982.1
U3 - shared alpha						981.6
U3 - dose-specific alpha						980.9

Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.20	0.20	0.20	0.20	0.20	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
MTD selection (%)						
U2 - shared alpha	0	1.7	1.2	9.7	87.4	0
U2 - dose-specific alpha	0	1.6	1	9	88.4	0
U3 - shared alpha	0	2.2	1.8	10.1	85.9	0
U3 - dose-specific alpha	0	2.3	1.3	10.6	85.8	0
OBD selection (%)						
U2 - shared alpha	40.6	24.4	14.6	13.1	7.3	0
U2 - dose-specific alpha	41	24.4	14.9	13.3	6.4	0
U3 - shared alpha	47.1	24.2	12.6	11	5.1	0
U3 - dose-specific alpha	45.3	24.8	13.4	11.9	4.6	0
Mean administrations						
U2 - shared alpha	4.5	4.9	6	8.3	6.3	-
U2 - dose-specific alpha	4.4	4.9	6	8.3	6.5	-
U3 - shared alpha	4.7	5.1	6.2	8.2	5.7	-
U3 - dose-specific alpha	4.7	5.1	6.2	8.3	5.7	-
Mean IPDE administrations						
U2 - shared alpha	0.6	2.9	2.2	3.6	2.3	-
U2 - dose-specific alpha	0.6	2.8	2.2	3.6	2.4	-
U3 - shared alpha	0.7	3	2.3	3.6	2.1	-
U3 - dose-specific alpha	0.7	3	2.3	3.6	2.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1009.6
U2 - dose-specific alpha						1010.1
U3 - shared alpha						1008.4
U3 - dose-specific alpha						1007.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.30	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
MTD selection (%)						
U2 - shared alpha	3.7	56	34.7	5.1	0.4	0.1
U2 - dose-specific alpha	4.5	55.4	34.3	5.3	0.4	0.1
U3 - shared alpha	4.1	60.5	31	4.1	0.2	0.1
U3 - dose-specific alpha	4.3	60.6	31	4	0	0.1
OBD selection (%)						
U2 - shared alpha	18.3	61	17.4	2.8	0.1	0.4
U2 - dose-specific alpha	19.4	59.5	17.7	2.7	0.1	0.6
U3 - shared alpha	23.2	61.2	13.2	1.9	0	0.5
U3 - dose-specific alpha	23.7	60.5	13.7	1.6	0	0.5
Mean administrations						
U2 - shared alpha	9	14.1	5.4	1.3	0.2	-
U2 - dose-specific alpha	8.9	14.2	5.3	1.3	0.2	-
U3 - shared alpha	9.5	14.5	4.8	1	0.1	-
U3 - dose-specific alpha	9.6	14.5	4.7	1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	1.9	4.7	1.4	0.4	0	-
U2 - dose-specific alpha	1.9	4.7	1.3	0.4	0	-
U3 - shared alpha	2.1	4.8	1.2	0.3	0	-
U3 - dose-specific alpha	2.2	4.8	1.2	0.3	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1190.1
U2 - dose-specific alpha						1189.2
U3 - shared alpha						1183.2
U3 - dose-specific alpha						1183.3

Scenario 18						
DLT rate	0.02	0.04	0.10	0.20	0.35	
Efficacy rate	0.05	0.10	0.15	0.20	0.50	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
MTD selection (%)						
U2 - shared alpha	0	0.2	1	21.8	77	0
U2 - dose-specific alpha	0	0.2	1.5	21	77.3	0
U3 - shared alpha	0	0.2	1.2	21.3	77.3	0
U3 - dose-specific alpha	0	0.2	1.5	21.5	76.8	0
OBD selection (%)						
U2 - shared alpha	10.6	10.8	8.3	15	55.3	0
U2 - dose-specific alpha	10.9	9.7	9.3	14.9	55.2	0
U3 - shared alpha	16.5	14.6	9.8	11.1	48	0
U3 - dose-specific alpha	17.9	13.9	9.4	11.5	47.3	0
Mean administrations						
U2 - shared alpha	3.4	4	5.1	6.8	10.6	-
U2 - dose-specific alpha	3.5	4	5.2	6.7	10.6	-
U3 - shared alpha	3.5	4.2	5.2	6.8	10.3	-
U3 - dose-specific alpha	3.6	4.2	5.2	6.8	10.2	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.8	1.8	2.8	3	-
U2 - dose-specific alpha	0.2	2.8	1.8	2.8	3	-
U3 - shared alpha	0.2	2.9	1.8	2.9	2.9	-
U3 - dose-specific alpha	0.3	2.9	1.8	2.8	2.9	-
Mean trial duration (days)						Days
U2 - shared alpha						1067
U2 - dose-specific alpha						1067
U3 - shared alpha						1062.6
U3 - dose-specific alpha						1061.9

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.60	0.77	
Efficacy rate	0.08	0.30	0.32	0.35	0.40	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
MTD selection (%)						
U2 - shared alpha	15	41.3	35.7	7.9	0.1	0
U2 - dose-specific alpha	14.7	41.6	36.2	7.3	0.2	0
U3 - shared alpha	14.6	42.5	34.9	7.9	0.1	0
U3 - dose-specific alpha	14.6	42.2	35.2	7.9	0.1	0
OBD selection (%)						
U2 - shared alpha	11	53.4	19.8	3	0.1	12.7
U2 - dose-specific alpha	11.1	53.5	19.9	2.7	0.2	12.6
U3 - shared alpha	13.5	55.2	16.6	2.2	0.1	12.4
U3 - dose-specific alpha	13.2	54.9	17	2.2	0.1	12.6
Mean administrations						
U2 - shared alpha	7.1	12.1	6.6	2.3	0.4	-
U2 - dose-specific alpha	7.1	12.2	6.6	2.3	0.4	-
U3 - shared alpha	7.4	12.5	6.3	2.1	0.4	-
U3 - dose-specific alpha	7.4	12.4	6.3	2.1	0.4	-
Mean IPDE administrations						
U2 - shared alpha	1.5	5.2	1.9	0.8	0.1	-
U2 - dose-specific alpha	1.4	5.3	1.9	0.8	0.1	-
U3 - shared alpha	1.5	5.4	1.8	0.7	0.1	-
U3 - dose-specific alpha	1.5	5.4	1.8	0.7	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1047.1
U2 - dose-specific alpha						1048.1
U3 - shared alpha						1047.4
U3 - dose-specific alpha						1043.6

Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
MTD selection (%)						
U2 - shared alpha	2	10	42.1	36.6	9.3	0
U2 - dose-specific alpha	1.8	10.6	43.2	35.3	9.1	0
U3 - shared alpha	1.7	11.1	41.3	36.8	9.1	0
U3 - dose-specific alpha	1.6	10.9	42.2	35.7	9.6	0
OBD selection (%)						
U2 - shared alpha	10.6	45.6	27.1	12.6	1.9	2.2
U2 - dose-specific alpha	11	46.5	26.6	11.7	2.2	2
U3 - shared alpha	12.4	50.1	22.6	11.5	1.3	2.1
U3 - dose-specific alpha	13.1	48.1	24.1	11.6	1.2	1.9
Mean administrations						
U2 - shared alpha	4.6	9.9	8.1	5.2	1.9	-
U2 - dose-specific alpha	4.7	9.8	8.2	5.2	1.8	-
U3 - shared alpha	4.9	10.4	7.8	4.9	1.7	-
U3 - dose-specific alpha	4.9	10.4	7.8	4.9	1.7	-
Mean IPDE administrations						
U2 - shared alpha	0.6	4.8	2.6	1.8	0.6	-
U2 - dose-specific alpha	0.7	4.8	2.6	1.8	0.6	-
U3 - shared alpha	0.8	5	2.5	1.7	0.6	-
U3 - dose-specific alpha	0.8	5	2.5	1.7	0.6	-
Mean trial duration (days)						Days
U2 - shared alpha						1059.4
U2 - dose-specific alpha						1059.1
U3 - shared alpha						1056.2
U3 - dose-specific alpha						1055.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.50	0.50	0.50	0.50	0.50	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
MTD selection (%)						
U2 - shared alpha	58.7	4.9	0	0	0	36.4
U2 - dose-specific alpha	58.8	4.9	0	0	0	36.3
U3 - shared alpha	57.7	4.9	0	0	0	37.4
U3 - dose-specific alpha	59.5	4.7	0	0	0	35.8
OBD selection (%)						
U2 - shared alpha	58.8	1.8	0	0	0	39.4
U2 - dose-specific alpha	58.7	2	0	0	0	39.3
U3 - shared alpha	59	1.6	0	0	0	39.4
U3 - dose-specific alpha	60	1.4	0	0	0	38.6
Mean administrations						
U2 - shared alpha	20.3	2.7	0.3	0	0	-
U2 - dose-specific alpha	20.3	2.6	0.3	0	0	-
U3 - shared alpha	20.4	2.6	0.2	0	0	-
U3 - dose-specific alpha	20.6	2.5	0.2	0	0	-
Mean IPDE administrations						
U2 - shared alpha	3.8	1	0.1	0	0	-
U2 - dose-specific alpha	3.8	1	0.1	0	0	-
U3 - shared alpha	3.9	1	0.1	0	0	-
U3 - dose-specific alpha	3.9	1	0	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1012.1
U2 - dose-specific alpha						1011
U3 - shared alpha						1009.4
U3 - dose-specific alpha						1013.8

Scenario 22						
DLT rate	0.07	0.27	0.50	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.50	0.50	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
MTD selection (%)						
U2 - shared alpha	12	49.3	33.5	5.2	0	0
U2 - dose-specific alpha	11.3	50.2	33.4	5.1	0	0
U3 - shared alpha	11.8	47.9	35.3	5	0	0
U3 - dose-specific alpha	12	47.9	35.1	5	0	0
OBD selection (%)						
U2 - shared alpha	14.4	56.9	13.6	4	0	11.1
U2 - dose-specific alpha	13.5	57.6	13.8	3.9	0	11.2
U3 - shared alpha	16.5	57.2	11.8	3.2	0	11.3
U3 - dose-specific alpha	16.1	57.8	11.2	3.3	0	11.6
Mean administrations						
U2 - shared alpha	7.8	12.9	5.8	2.2	0.2	-
U2 - dose-specific alpha	7.9	12.9	5.8	2.2	0.2	-
U3 - shared alpha	8.2	13	5.5	2.1	0.2	-
U3 - dose-specific alpha	8.2	12.9	5.6	2	0.2	-
Mean IPDE administrations						
U2 - shared alpha	1.8	5.7	1.6	0.7	0.1	-
U2 - dose-specific alpha	1.8	5.7	1.6	0.7	0.1	-
U3 - shared alpha	2	5.7	1.6	0.7	0.1	-
U3 - dose-specific alpha	2	5.7	1.6	0.7	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1045.6
U2 - dose-specific alpha						1046.2
U3 - shared alpha						1039.7
U3 - dose-specific alpha						1039.5

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 23						
DLT rate	0.32	0.55	0.80	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
MTD selection (%)						
U2 - shared alpha	74.9	6.4	0.2	0	0	18.5
U2 - dose-specific alpha	74.9	6.2	0.3	0	0	18.6
U3 - shared alpha	75.2	6	0.3	0	0	18.5
U3 - dose-specific alpha	74.3	6.5	0.3	0	0	18.9
OBD selection (%)						
U2 - shared alpha	8.1	2	0	0	0	89.9
U2 - dose-specific alpha	8.3	2.1	0.1	0	0	89.5
U3 - shared alpha	8.3	1.7	0.1	0	0	89.9
U3 - dose-specific alpha	8.3	1.9	0.1	0	0	89.7
Mean administrations						
U2 - shared alpha	11.1	4.4	0.7	0.1	0	-
U2 - dose-specific alpha	11.1	4.4	0.7	0.1	0	-
U3 - shared alpha	11.1	4.4	0.7	0.1	0	-
U3 - dose-specific alpha	11.2	4.4	0.7	0.1	0	-
Mean IPDE administrations						
U2 - shared alpha	2.9	2.3	0.2	0	0	-
U2 - dose-specific alpha	2.9	2.3	0.2	0	0	-
U3 - shared alpha	2.8	2.3	0.2	0	0	-
U3 - dose-specific alpha	2.9	2.3	0.2	0	0	-
Mean trial duration (days)						Days
U2 - shared alpha						594.5
U2 - dose-specific alpha						594.3
U3 - shared alpha						591.5
U3 - dose-specific alpha						594.6

Scenario 24						
DLT rate	0.01	0.02	0.05	0.10	0.25	
Efficacy rate	0.05	0.08	0.20	0.50	0.50	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
MTD selection (%)						
U2 - shared alpha	0	0.1	0	26.8	73.1	0
U2 - dose-specific alpha	0	0.3	0	27.2	72.5	0
U3 - shared alpha	0	0.1	0.1	30.7	69.1	0
U3 - dose-specific alpha	0	0.1	0	30.8	69.1	0
OBD selection (%)						
U2 - shared alpha	2.2	2.2	3.6	54.1	37.9	0
U2 - dose-specific alpha	2.4	1.4	3.4	55.2	37.5	0.1
U3 - shared alpha	4	3.1	4.8	57.1	31	0
U3 - dose-specific alpha	4	3.1	5.3	57.7	29.9	0
Mean administrations						
U2 - shared alpha	3.1	3.1	3.9	12.5	7.4	-
U2 - dose-specific alpha	3.1	3.1	3.9	12.4	7.5	-
U3 - shared alpha	3.1	3.2	4	13	6.7	-
U3 - dose-specific alpha	3.1	3.2	4	12.9	6.8	-
Mean IPDE administrations						
U2 - shared alpha	0	2.4	1.2	4.4	2.2	-
U2 - dose-specific alpha	0.1	2.4	1.2	4.3	2.2	-
U3 - shared alpha	0	2.5	1.3	4.6	2	-
U3 - dose-specific alpha	0	2.4	1.2	4.5	2	-
Mean trial duration (days)						Days
U2 - shared alpha						1090.7
U2 - dose-specific alpha						1089.4
U3 - shared alpha						1087.4
U3 - dose-specific alpha						1089.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.20	0.25	0.40	0.45	0.50	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
MTD selection (%)						
U2 - shared alpha	2	30.3	55.4	11.7	0.6	0
U2 - dose-specific alpha	2.5	31.4	53.8	11.9	0.4	0
U3 - shared alpha	2.1	31.6	53.3	12.6	0.4	0
U3 - dose-specific alpha	2	30.9	54.5	12.3	0.3	0
OBD selection (%)						
U2 - shared alpha	25.6	29.7	37.1	6.3	0.1	1.2
U2 - dose-specific alpha	26.8	29	35.8	7.1	0.1	1.2
U3 - shared alpha	31.9	29.3	31.9	5.6	0	1.3
U3 - dose-specific alpha	31.5	29.2	32	6	0	1.3
Mean administrations						
U2 - shared alpha	8.7	9	8.8	2.9	0.4	-
U2 - dose-specific alpha	8.8	9.1	8.7	2.9	0.4	-
U3 - shared alpha	9.4	9.1	8.5	2.5	0.3	-
U3 - dose-specific alpha	9.3	9.1	8.5	2.6	0.4	-
Mean IPDE administrations						
U2 - shared alpha	2.2	4	2.4	0.9	0.1	-
U2 - dose-specific alpha	2.2	4	2.3	0.9	0.1	-
U3 - shared alpha	2.5	4	2.3	0.8	0.1	-
U3 - dose-specific alpha	2.4	4.1	2.3	0.8	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1121.3
U2 - dose-specific alpha						1119.8
U3 - shared alpha						1114.8
U3 - dose-specific alpha						1115.1

Scenario 26						
DLT rate	0.15	0.22	0.30	0.62	0.80	
Efficacy rate	0.10	0.20	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
MTD selection (%)						
U2 - shared alpha	11.3	30.8	43.5	13.8	0.3	0.3
U2 - dose-specific alpha	12	30.2	43.3	13.9	0.3	0.3
U3 - shared alpha	11.8	29.9	42.3	15.5	0.2	0.3
U3 - dose-specific alpha	11.6	30.6	42.8	14.4	0.3	0.3
OBD selection (%)						
U2 - shared alpha	18.7	29.8	32.9	7.8	0.1	10.7
U2 - dose-specific alpha	19.2	28.2	33.6	7.6	0.1	11.3
U3 - shared alpha	21.6	28.7	31	7.3	0.1	11.3
U3 - dose-specific alpha	23	30.6	28.3	7.1	0.1	10.9
Mean administrations						
U2 - shared alpha	8	8.6	8.3	3.2	0.4	-
U2 - dose-specific alpha	8	8.7	8.3	3.2	0.4	-
U3 - shared alpha	8.3	8.7	8.2	3	0.4	-
U3 - dose-specific alpha	8.4	8.8	8.1	2.9	0.4	-
Mean IPDE administrations						
U2 - shared alpha	1.9	4.1	2.3	1.1	0.1	-
U2 - dose-specific alpha	1.9	4.1	2.3	1.1	0.1	-
U3 - shared alpha	2	4.2	2.3	1	0.1	-
U3 - dose-specific alpha	2.1	4.2	2.2	1	0.1	-
Mean trial duration (days)						Days
U2 - shared alpha						1045.7
U2 - dose-specific alpha						1045.7
U3 - shared alpha						1041.9
U3 - dose-specific alpha						1042.5

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 27						
DLT rate	0.11	0.17	0.30	0.50	0.62	
Efficacy rate	0.10	0.20	0.40	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
MTD selection (%)						
U2 - shared alpha	4.4	17.5	51.6	23.9	2.6	0
U2 - dose-specific alpha	4	17.9	52.2	23.7	2.2	0
U3 - shared alpha	4.4	17.4	52.1	24	2.1	0
U3 - dose-specific alpha	4.3	17.7	52	23.8	2.2	0
OBD selection (%)						
U2 - shared alpha	12.9	23.4	46.3	11.8	0.7	4.9
U2 - dose-specific alpha	12.7	22.9	47.8	11.4	0.6	4.6
U3 - shared alpha	15.8	25.7	42.3	11	0.5	4.7
U3 - dose-specific alpha	16.7	25.4	42	10.4	0.7	4.8
Mean administrations						
U2 - shared alpha	6.2	8	10.3	4.1	0.9	-
U2 - dose-specific alpha	6.2	8	10.3	4	0.9	-
U3 - shared alpha	6.5	8.2	10.1	3.9	0.7	-
U3 - dose-specific alpha	6.5	8.2	10.1	3.9	0.8	-
Mean IPDE administrations						
U2 - shared alpha	1.2	3.9	2.9	1.3	0.3	-
U2 - dose-specific alpha	1.3	3.9	2.9	1.3	0.3	-
U3 - shared alpha	1.4	4	2.9	1.3	0.2	-
U3 - dose-specific alpha	1.4	4	2.9	1.2	0.2	-
Mean trial duration (days)						Days
U2 - shared alpha						1087.9
U2 - dose-specific alpha						1089.9
U3 - shared alpha						1085.5
U3 - dose-specific alpha						1084.5

Scenario 28						
DLT rate	0.20	0.30	0.50	0.60	0.77	
Efficacy rate	0.05	0.30	0.32	0.35	0.40	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
MTD selection (%)						
U2 - shared alpha	39.2	39.3	17.8	2	0	1.7
U2 - dose-specific alpha	40.2	39.5	16.6	2.1	0	1.6
U3 - shared alpha	40	39	17.6	1.7	0	1.7
U3 - dose-specific alpha	39.7	38.3	18.6	1.7	0	1.7
OBD selection (%)						
U2 - shared alpha	9	44.6	9.1	0.6	0	36.7
U2 - dose-specific alpha	9.4	44.5	8.6	0.5	0	37
U3 - shared alpha	8.9	45.4	7.9	0.1	0	37.7
U3 - dose-specific alpha	9.5	44.8	8.6	0.2	0	36.9
Mean administrations						
U2 - shared alpha	8.5	11.2	4.1	1.2	0.2	-
U2 - dose-specific alpha	8.5	11.1	4.1	1.2	0.2	-
U3 - shared alpha	8.7	11.2	4	1.1	0.1	-
U3 - dose-specific alpha	8.7	11.3	3.9	1.1	0.1	-
Mean IPDE administrations						
U2 - shared alpha	1.9	4.8	1.2	0.4	0	-
U2 - dose-specific alpha	1.9	4.7	1.2	0.4	0	-
U3 - shared alpha	2	4.8	1.2	0.4	0	-
U3 - dose-specific alpha	2	4.8	1.1	0.4	0	-
Mean trial duration (days)						Days
U2 - shared alpha						928
U2 - dose-specific alpha						925.2
U3 - shared alpha						921.7
U3 - dose-specific alpha						920.7

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 29						
DLT rate	0.19	0.30	0.56	0.74	0.86	
Efficacy rate	0.40	0.40	0.40	0.40	0.40	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
MTD selection (%)						
U2 - shared alpha	23.4	60.3	13.5	0.7	0	2.1
U2 - dose-specific alpha	23.7	60.7	12.6	0.9	0	2.1
U3 - shared alpha	23.9	61.2	12.1	0.7	0	2.1
U3 - dose-specific alpha	23.8	61.2	12.2	0.8	0	2
OBD selection (%)						
U2 - shared alpha	63.1	30	4.4	0.2	0	2.3
U2 - dose-specific alpha	63.1	29.9	4.4	0.2	0	2.4
U3 - shared alpha	67.2	27.3	3	0.2	0	2.3
U3 - dose-specific alpha	68.2	26.5	3	0.2	0	2.1
Mean administrations						
U2 - shared alpha	17.3	8.9	2.7	0.6	0	-
U2 - dose-specific alpha	17.4	8.9	2.6	0.6	0	-
U3 - shared alpha	18.2	8.6	2.2	0.5	0	-
U3 - dose-specific alpha	18.4	8.5	2.2	0.5	0	-
Mean IPDE administrations						
U2 - shared alpha	4.2	3.1	0.7	0.2	0	-
U2 - dose-specific alpha	4.2	3.1	0.7	0.2	0	-
U3 - shared alpha	4.6	3	0.6	0.1	0	-
U3 - dose-specific alpha	4.6	3	0.6	0.1	0	-
Mean trial duration (days)						Days
U2 - shared alpha						1177.9
U2 - dose-specific alpha						1177.7
U3 - shared alpha						1172.2
U3 - dose-specific alpha						1173.2

Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.20	0.40	0.50	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
MTD selection (%)						
U2 - shared alpha	0.4	1.8	12.3	53.6	31.9	0
U2 - dose-specific alpha	0.4	1.5	12.4	55.7	30	0
U3 - shared alpha	0.4	1.5	12.3	56.1	29.7	0
U3 - dose-specific alpha	0.4	1.4	12.3	57.4	28.5	0
OBD selection (%)						
U2 - shared alpha	17.2	12.2	12.2	43.7	14.3	0.4
U2 - dose-specific alpha	17.4	12.1	12.4	43.7	14	0.4
U3 - shared alpha	25.6	14.5	10.1	38.4	10.9	0.5
U3 - dose-specific alpha	25.2	14.7	10.6	39.2	9.8	0.5
Mean administrations						
U2 - shared alpha	4.3	4.8	5.9	10.4	4.6	-
U2 - dose-specific alpha	4.3	4.8	5.9	10.3	4.7	-
U3 - shared alpha	4.5	5	6	10.3	4.3	-
U3 - dose-specific alpha	4.5	4.9	6	10.3	4.2	-
Mean IPDE administrations						
U2 - shared alpha	0.6	2.9	1.9	3.4	1.3	-
U2 - dose-specific alpha	0.5	2.9	1.9	3.4	1.3	-
U3 - shared alpha	0.6	3	1.9	3.4	1.2	-
U3 - dose-specific alpha	0.6	3	1.9	3.4	1.2	-
Mean trial duration (days)						Days
U2 - shared alpha						1096.3
U2 - dose-specific alpha						1095.4
U3 - shared alpha						1091.3
U3 - dose-specific alpha						1091.7

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 31						
DLT rate	0.04	0.06	0.10	0.20	0.60	
Efficacy rate	0.20	0.50	0.50	0.50	0.50	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
MTD selection (%)						
U2 - shared alpha	0	21.4	24.2	33.6	20.8	0
U2 - dose-specific alpha	0	21.5	23.2	34	21.3	0
U3 - shared alpha	0	26.8	24.2	32.1	16.9	0
U3 - dose-specific alpha	0	26	24.9	31.6	17.5	0
OBD selection (%)						
U2 - shared alpha	5.3	34.7	27	25.9	7.1	0
U2 - dose-specific alpha	5.2	33.9	27.2	26.3	7.4	0
U3 - shared alpha	6.8	41.2	26.4	21.2	4.4	0
U3 - dose-specific alpha	6.8	40.1	26.5	21.3	5.3	0
Mean administrations						
U2 - shared alpha	3.8	11	7.4	5.9	2	-
U2 - dose-specific alpha	3.8	10.9	7.4	5.9	2.1	-
U3 - shared alpha	3.9	11.9	7.2	5.4	1.6	-
U3 - dose-specific alpha	3.9	11.8	7.3	5.4	1.6	-
Mean IPDE administrations						
U2 - shared alpha	0.3	4.4	2.2	1.7	0.6	-
U2 - dose-specific alpha	0.3	4.4	2.2	1.7	0.6	-
U3 - shared alpha	0.3	4.7	2.1	1.6	0.5	-
U3 - dose-specific alpha	0.3	4.7	2.2	1.6	0.5	-
Mean trial duration (days)						Days
U2 - shared alpha						1153.1
U2 - dose-specific alpha						1153.1
U3 - shared alpha						1147.9
U3 - dose-specific alpha						1147.5

Scenario 32						
DLT rate	0.02	0.05	0.11	0.30	0.36	
Efficacy rate	0.08	0.15	0.30	0.30	0.30	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
MTD selection (%)						
U2 - shared alpha	0.1	0.2	6.5	35	58.2	0
U2 - dose-specific alpha	0.1	0.2	5.9	34.7	59.1	0
U3 - shared alpha	0.1	0.2	5.4	39.8	54.5	0
U3 - dose-specific alpha	0.1	0.2	5.9	38.3	55.5	0
OBD selection (%)						
U2 - shared alpha	13.6	14.2	34.4	25.7	12	0.1
U2 - dose-specific alpha	14.4	13.6	34.1	25.7	12.1	0.1
U3 - shared alpha	20	16.5	32.5	23.6	7.3	0.1
U3 - dose-specific alpha	19.3	17.7	31.8	23.4	7.7	0.1
Mean administrations						
U2 - shared alpha	3.5	4.3	8.5	8.4	5.3	-
U2 - dose-specific alpha	3.5	4.2	8.5	8.5	5.3	-
U3 - shared alpha	3.6	4.5	8.8	8.4	4.7	-
U3 - dose-specific alpha	3.6	4.5	8.7	8.5	4.7	-
Mean IPDE administrations						
U2 - shared alpha	0.2	2.9	2.8	3.2	1.7	-
U2 - dose-specific alpha	0.2	2.8	2.8	3.2	1.7	-
U3 - shared alpha	0.3	2.9	2.9	3.2	1.5	-
U3 - dose-specific alpha	0.3	2.9	2.9	3.2	1.5	-
Mean trial duration (days)						Days
U2 - shared alpha						1057.6
U2 - dose-specific alpha						1058.6
U3 - shared alpha						1054.2
U3 - dose-specific alpha						1054.7

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.20	0.40	0.50	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
MTD selection (%)						
U2 - shared alpha	0	0.1	5.7	43.5	50.7	0
U2 - dose-specific alpha	0	0.2	6	43.1	50.7	0
U3 - shared alpha	0	0.1	5.6	44.4	49.9	0
U3 - dose-specific alpha	0	0.2	6	44.5	49.3	0
OBD selection (%)						
U2 - shared alpha	7.9	5.6	12.1	43.7	30.6	0.1
U2 - dose-specific alpha	8.6	5.3	12.1	43.7	30.1	0.2
U3 - shared alpha	12	7.6	11	43.6	25.7	0.1
U3 - dose-specific alpha	12.2	8.2	11.6	42.5	25.4	0.1
Mean administrations						
U2 - shared alpha	3.3	3.6	5.9	10.5	6.7	-
U2 - dose-specific alpha	3.3	3.6	5.9	10.3	6.8	-
U3 - shared alpha	3.4	3.8	6	10.6	6.2	-
U3 - dose-specific alpha	3.4	3.7	6	10.5	6.3	-
Mean IPDE administrations						
U2 - shared alpha	0.1	2.7	1.8	3.7	1.8	-
U2 - dose-specific alpha	0.2	2.6	1.8	3.7	1.8	-
U3 - shared alpha	0.2	2.7	1.8	3.8	1.6	-
U3 - dose-specific alpha	0.2	2.7	1.8	3.8	1.7	-
Mean trial duration (days)						Days
U2 - shared alpha						1094.7
U2 - dose-specific alpha						1093.3
U3 - shared alpha						1090.6
U3 - dose-specific alpha						1091.5

Scenario 34						
DLT rate	0.05	0.13	0.30	0.36	0.42	
Efficacy rate	0.05	0.10	0.30	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
MTD selection (%)						
U2 - shared alpha	0.8	8.4	34.6	39.7	16.5	0
U2 - dose-specific alpha	0.8	8.7	35.1	38.6	16.8	0
U3 - shared alpha	0.6	8.2	34.8	42	14.4	0
U3 - dose-specific alpha	0.8	7.3	35.4	42.3	14.2	0
OBD selection (%)						
U2 - shared alpha	11.5	12.3	44.5	23	5.1	3.6
U2 - dose-specific alpha	12.4	12.2	43.8	22.8	5.2	3.6
U3 - shared alpha	15.8	14.3	42.1	20.9	3.3	3.6
U3 - dose-specific alpha	15.7	15.1	41.8	21.3	2.8	3.3
Mean administrations						
U2 - shared alpha	5	6.2	9.6	6.3	2.8	-
U2 - dose-specific alpha	4.9	6.1	9.6	6.3	2.8	-
U3 - shared alpha	5.2	6.3	9.7	6.1	2.4	-
U3 - dose-specific alpha	5.2	6.4	9.6	6.2	2.4	-
Mean IPDE administrations						
U2 - shared alpha	0.8	3.5	2.9	2.2	0.9	-
U2 - dose-specific alpha	0.8	3.5	2.9	2.2	0.9	-
U3 - shared alpha	1	3.6	2.9	2.1	0.8	-
U3 - dose-specific alpha	1	3.6	2.9	2.1	0.8	-
Mean trial duration (days)						Days
U2 - shared alpha						1065.3
U2 - dose-specific alpha						1066
U3 - shared alpha						1062.1
U3 - dose-specific alpha						1062.6

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 35						
DLT rate	0.03	0.09	0.30	0.40	0.45	
Efficacy rate	0.20	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
MTD selection (%)						
U2 - shared alpha	0.1	17.9	40.7	32.7	8.6	0
U2 - dose-specific alpha	0.1	17.8	41.5	31.9	8.7	0
U3 - shared alpha	0.1	20.4	43.4	28.6	7.5	0
U3 - dose-specific alpha	0.1	20.2	44	28	7.7	0
OBD selection (%)						
U2 - shared alpha	9.2	40	31.6	15.2	3.9	0.1
U2 - dose-specific alpha	9.1	39.7	32.5	15	3.6	0.1
U3 - shared alpha	13.6	45.6	27.4	10.6	2.7	0.1
U3 - dose-specific alpha	14	45.1	27.6	10.7	2.5	0.1
Mean administrations						
U2 - shared alpha	4.4	11.1	8.4	4.5	1.5	-
U2 - dose-specific alpha	4.4	10.9	8.5	4.6	1.6	-
U3 - shared alpha	4.6	12	8.1	4	1.3	-
U3 - dose-specific alpha	4.6	12	8.1	4.1	1.2	-
Mean IPDE administrations						
U2 - shared alpha	0.5	4.4	2.3	1.3	0.4	-
U2 - dose-specific alpha	0.5	4.4	2.4	1.3	0.5	-
U3 - shared alpha	0.6	4.8	2.2	1.2	0.3	-
U3 - dose-specific alpha	0.6	4.8	2.2	1.2	0.3	-
Mean trial duration (days)						Days
U2 - shared alpha						1154.1
U2 - dose-specific alpha						1155.2
U3 - shared alpha						1147.4
U3 - dose-specific alpha						1148.8

Scenario 36						
DLT rate	0.13	0.29	0.35	0.40	0.43	
Efficacy rate	0.05	0.20	0.22	0.30	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
MTD selection (%)						
U2 - shared alpha	18.7	31.8	28	15.6	5.7	0.2
U2 - dose-specific alpha	18.6	32	27.6	16.4	5.2	0.2
U3 - shared alpha	19.1	31	28.5	16.4	4.8	0.2
U3 - dose-specific alpha	19.1	30.8	28.8	15.9	5.2	0.2
OBD selection (%)						
U2 - shared alpha	12.2	41.6	16.1	7.9	1.7	20.5
U2 - dose-specific alpha	12	41.2	16.1	8.2	1.6	20.9
U3 - shared alpha	16	43.1	11.6	7.6	0.8	20.9
U3 - dose-specific alpha	16.1	41.3	13	7.1	1.1	21.4
Mean administrations						
U2 - shared alpha	7.5	9.8	5.6	3.6	1.2	-
U2 - dose-specific alpha	7.5	9.8	5.6	3.5	1.2	-
U3 - shared alpha	7.8	10	5.4	3.4	1.1	-
U3 - dose-specific alpha	7.7	10	5.4	3.4	1.1	-
Mean IPDE administrations						
U2 - shared alpha	1.6	4.8	1.7	1.3	0.4	-
U2 - dose-specific alpha	1.6	4.8	1.7	1.3	0.4	-
U3 - shared alpha	1.8	4.8	1.7	1.2	0.3	-
U3 - dose-specific alpha	1.7	4.9	1.7	1.2	0.4	-
Mean trial duration (days)						Days
U2 - shared alpha						976.9
U2 - dose-specific alpha						978.2
U3 - shared alpha						974.4
U3 - dose-specific alpha						973.3

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.

Phase I/II Operating Characteristics

Shared vs dose-specific additive-alpha efficacy - IPDE alpha = 0 - allocation = one-stage - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No MTD/OBD %
Scenario 37						
DLT rate	0.08	0.30	0.40	0.45	0.50	
Efficacy rate	0.05	0.20	0.20	0.20	0.20	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
MTD selection (%)						
U2 - shared alpha	15.2	37.7	32.8	11.5	2.8	0
U2 - dose-specific alpha	15.6	36.2	34.1	11	3.1	0
U3 - shared alpha	15.5	36.7	33.2	11.5	3.1	0
U3 - dose-specific alpha	16.1	35.6	33.5	12	2.8	0
OBD selection (%)						
U2 - shared alpha	12.2	48.8	16.7	4.1	0.8	17.4
U2 - dose-specific alpha	13.1	47.9	16.3	3.9	0.8	18
U3 - shared alpha	16	50.2	13.5	2.6	0.5	17.2
U3 - dose-specific alpha	17.1	48.7	14	2.4	0.6	17.2
Mean administrations						
U2 - shared alpha	7.4	11.2	5.8	3	0.9	-
U2 - dose-specific alpha	7.5	11.1	5.7	3.1	0.9	-
U3 - shared alpha	7.8	11.2	5.6	2.9	0.8	-
U3 - dose-specific alpha	7.8	11.2	5.5	2.9	0.8	-
Mean IPDE administrations						
U2 - shared alpha	1.6	5.4	1.8	1.1	0.3	-
U2 - dose-specific alpha	1.6	5.3	1.8	1.1	0.3	-
U3 - shared alpha	1.8	5.3	1.7	1.1	0.3	-
U3 - dose-specific alpha	1.8	5.4	1.7	1.1	0.3	-
Mean trial duration (days)						Days
U2 - shared alpha						993.5
U2 - dose-specific alpha						993.4
U3 - shared alpha						992.2
U3 - dose-specific alpha						992.2

Priors: efficacy beta-binomial Beta(0.5, 0.5); efficacy additive alpha Beta(0.15, 0.85); toxicity additive alpha Beta(0.15, 0.85).

Final column: early stopping % for MTD selection and no-OBD % for OBD selection; shaded rows are the dose-specific alpha efficacy model.